

(12) **United States Patent**
Choi et al.

(10) **Patent No.:** **US 12,385,499 B2**
(45) **Date of Patent:** **Aug. 12, 2025**

(54) **BLOWER**

(71) Applicant: **LG ELECTRONICS INC.**, Seoul
(KR)

(72) Inventors: **Jinwook Choi**, Seoul (KR); **Jaehyuk Jung**, Seoul (KR); **Juhyun Kim**, Seoul (KR); **Seokho Choi**, Seoul (KR)

(73) Assignee: **LG ELECTRONICS INC.**, Seoul
(KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/373,562**

(22) Filed: **Sep. 27, 2023**

(65) **Prior Publication Data**

US 2024/0151244 A1 May 9, 2024

(30) **Foreign Application Priority Data**

Nov. 4, 2022 (KR) 10-2022-0146104

(51) **Int. Cl.**

F04D 29/44 (2006.01)

F04D 17/00 (2006.01)

(52) **U.S. Cl.**

CPC **F04D 29/444** (2013.01); **F04D 17/00** (2013.01)

(58) **Field of Classification Search**

CPC F04D 29/444; F04D 17/00; F04D 17/06; F04D 29/4253; F04D 29/602; F04D 25/08; F04D 29/263; F04D 29/281; F04D 29/325; F04D 29/403; F04D 29/4206; F04D 29/441; F04D 29/522; F04D 29/541; F05D 2210/12

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2021/0277911 A1* 9/2021 Jung F04F 5/16

FOREIGN PATENT DOCUMENTS

CN 107023884 8/2017

CN 111156623 5/2020

CN 212838644 U * 3/2021

* cited by examiner

Primary Examiner — Courtney D Heinle

Assistant Examiner — Eric A Lange

(74) *Attorney, Agent, or Firm* — KED & ASSOCIATES

(57) **ABSTRACT**

A blower including a case; a fan; and a diffuser that is disposed downstream of the fan and guides air, discharged from the fan, toward discharge ports. The diffuser includes an outer wall; an inner wall spaced apart inwardly from the outer wall so that an air passage is formed therebetween; and a plurality of vanes that extends radially between the outer wall and the inner wall and divides the air passage into a plurality of unit passages. The outer wall and the inner wall are formed to have areas with different radial separation distances between the outer wall and the inner wall, such that in an area of the air passage of the diffuser in which air is discharged to a portion having a high flow resistance in the blower, the radial separation distance increases to widen the air passage, thereby compensating for asymmetry of the passage inside of the blower, allowing a uniform volume of air to reach an entire area of the discharge ports, and improving performance of the blower.

20 Claims, 17 Drawing Sheets

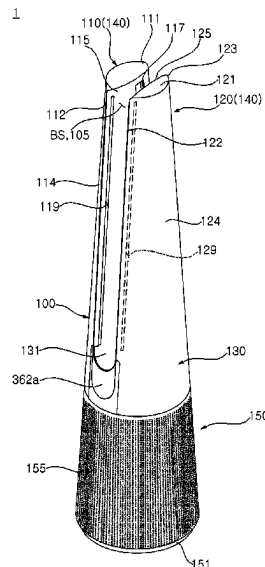


FIG. 1

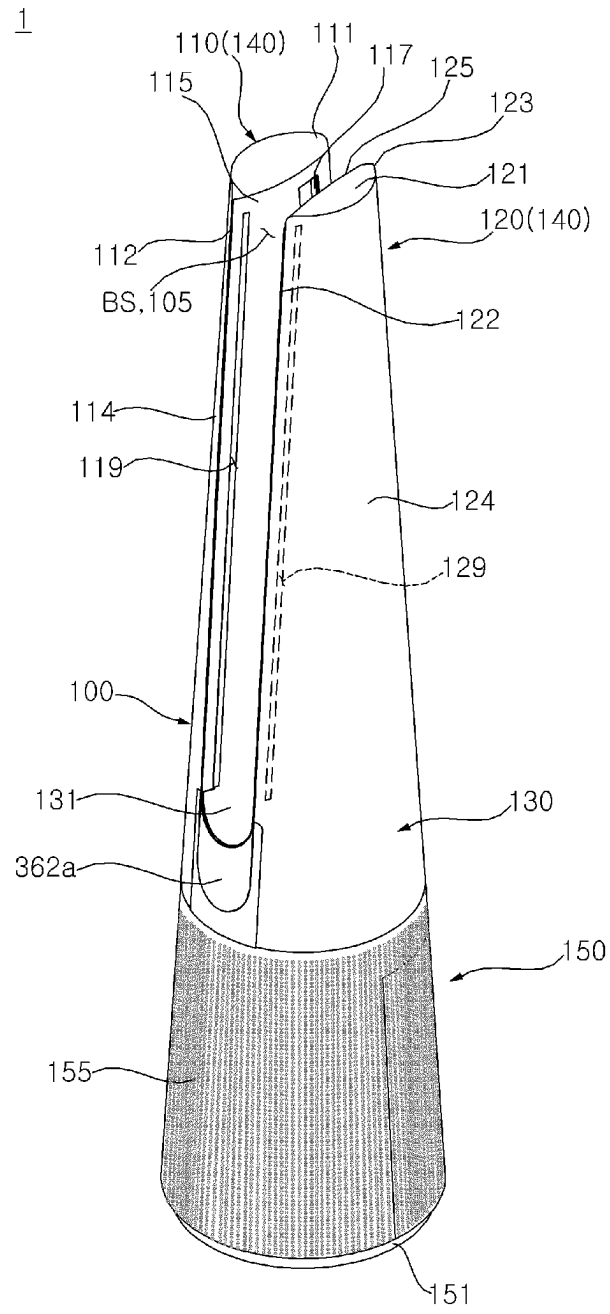


FIG. 2

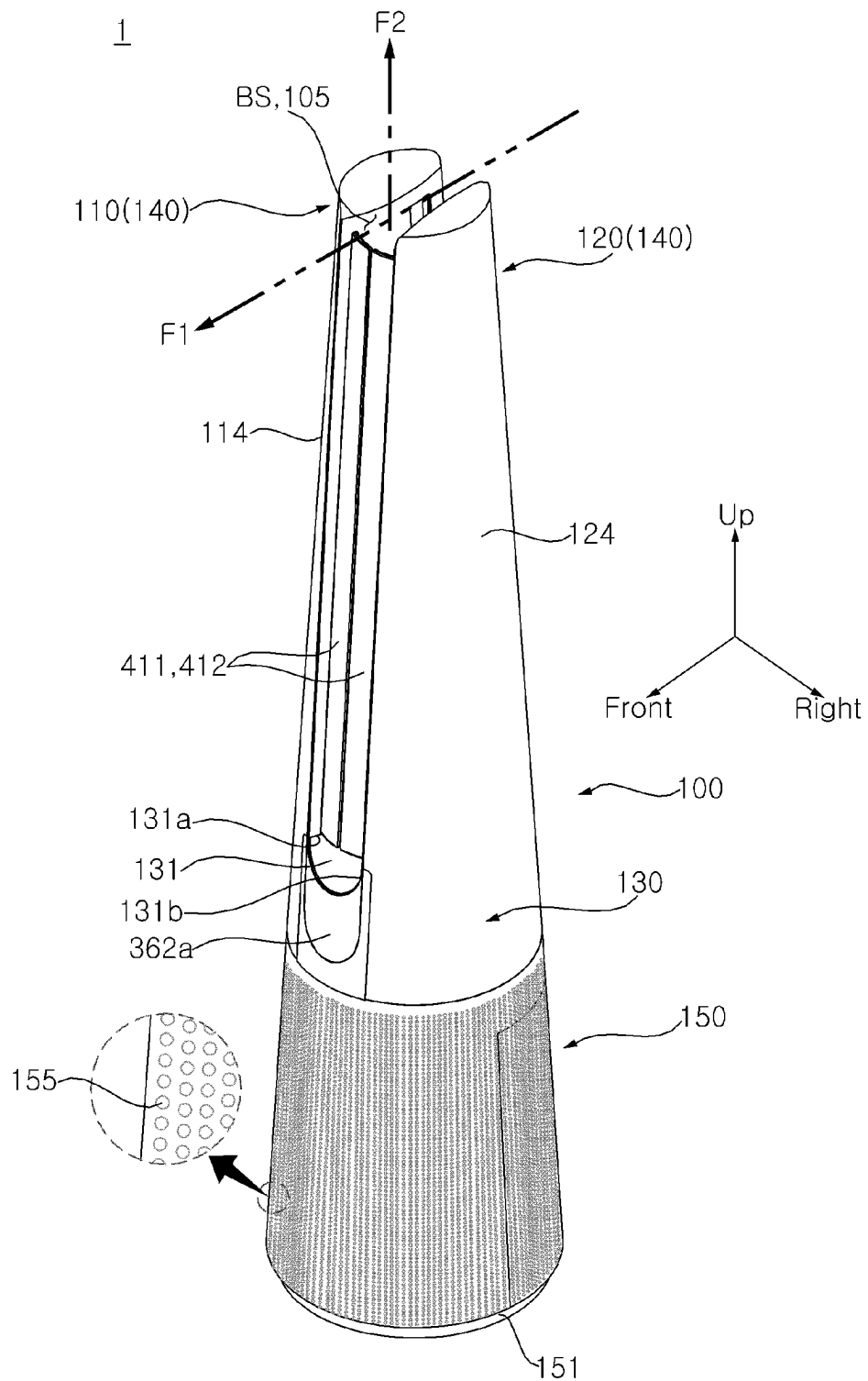


FIG. 3

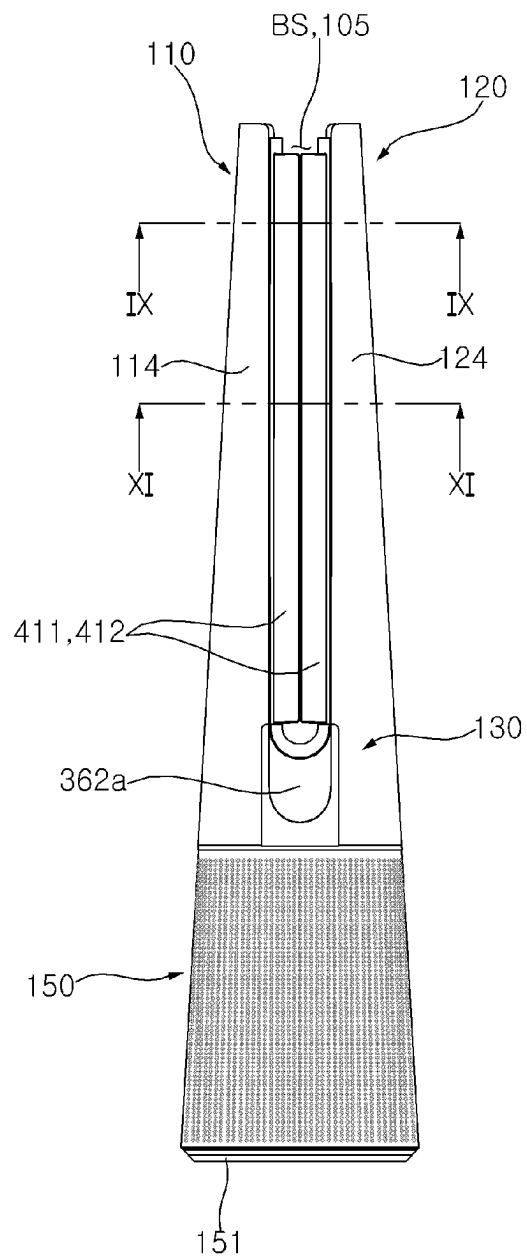


FIG. 4

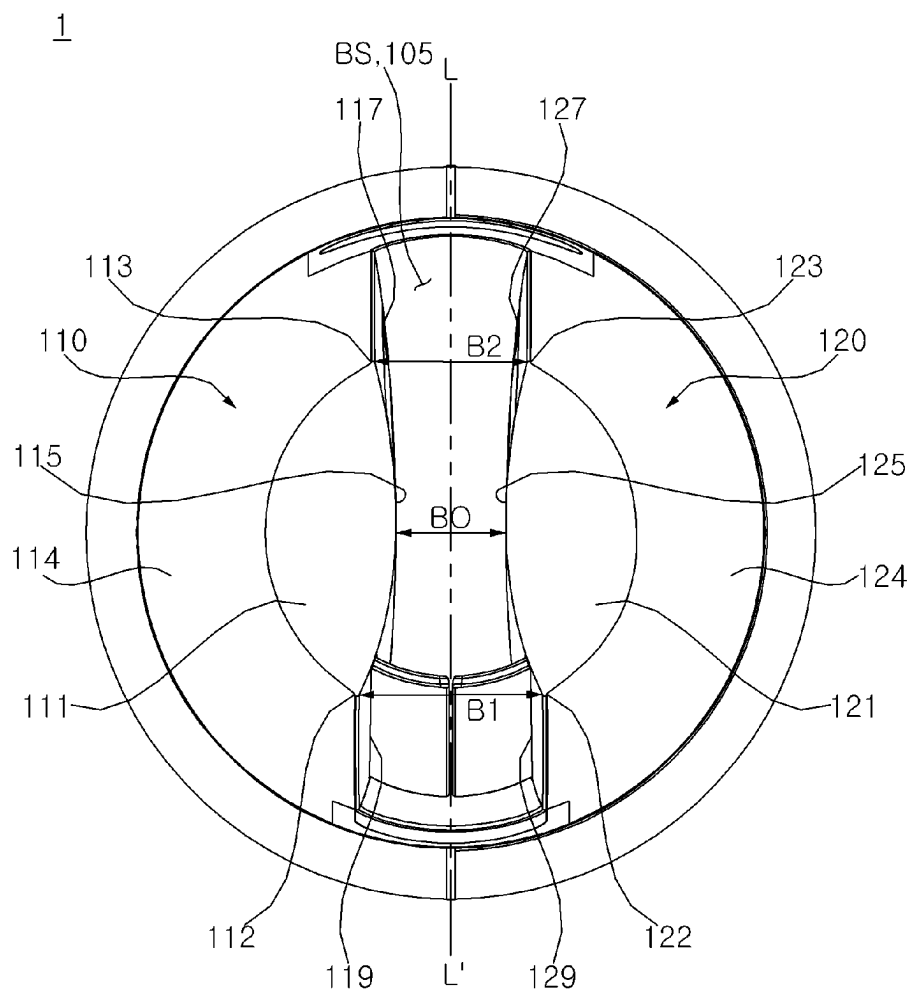


FIG. 5

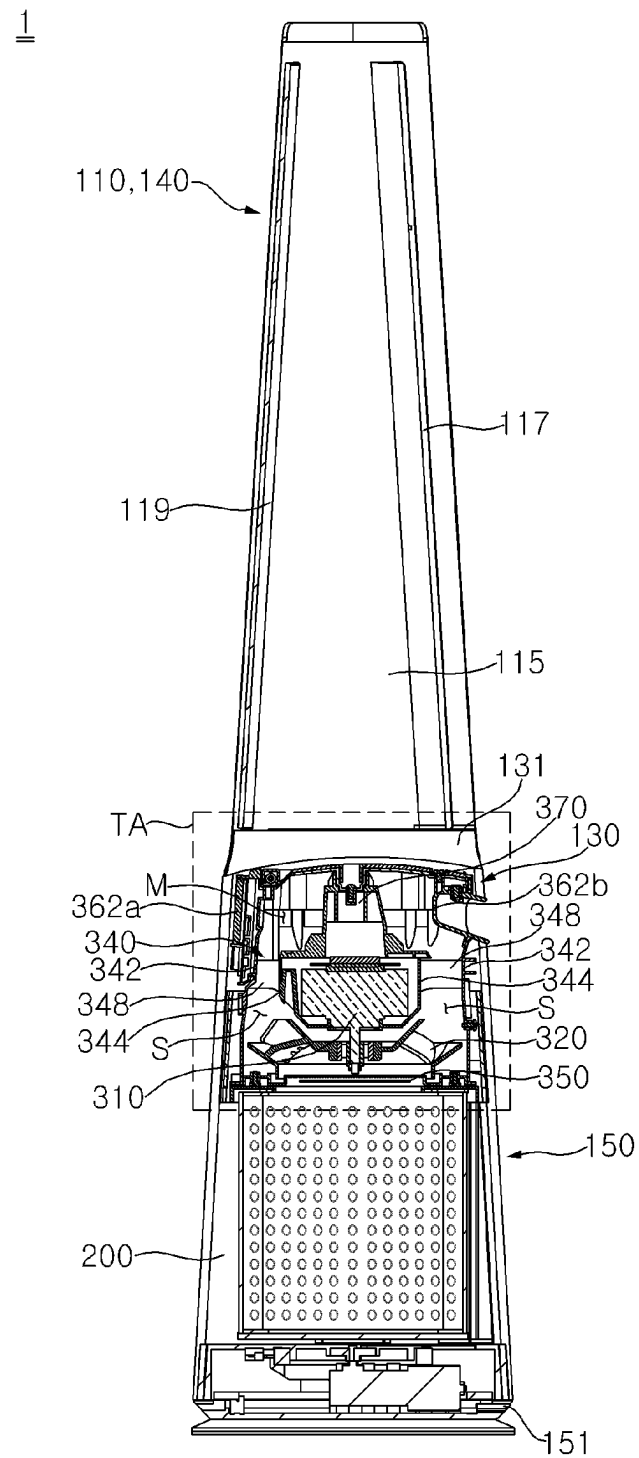


FIG. 6

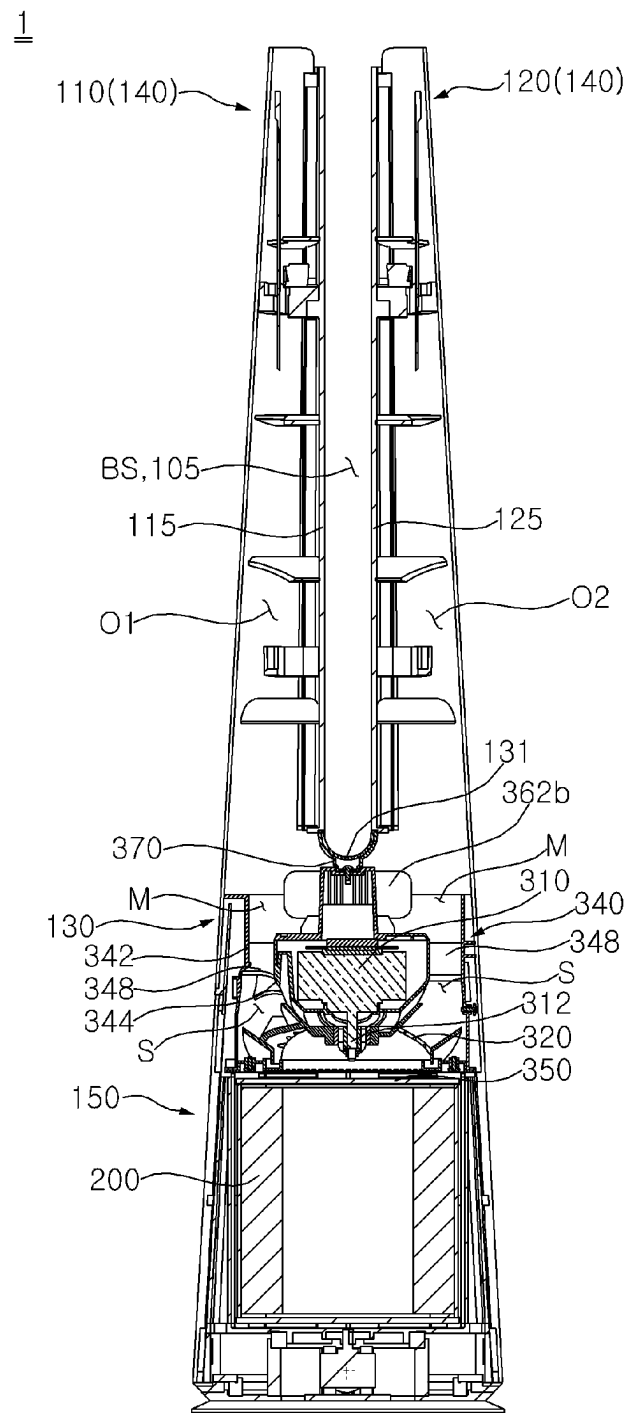


FIG. 7

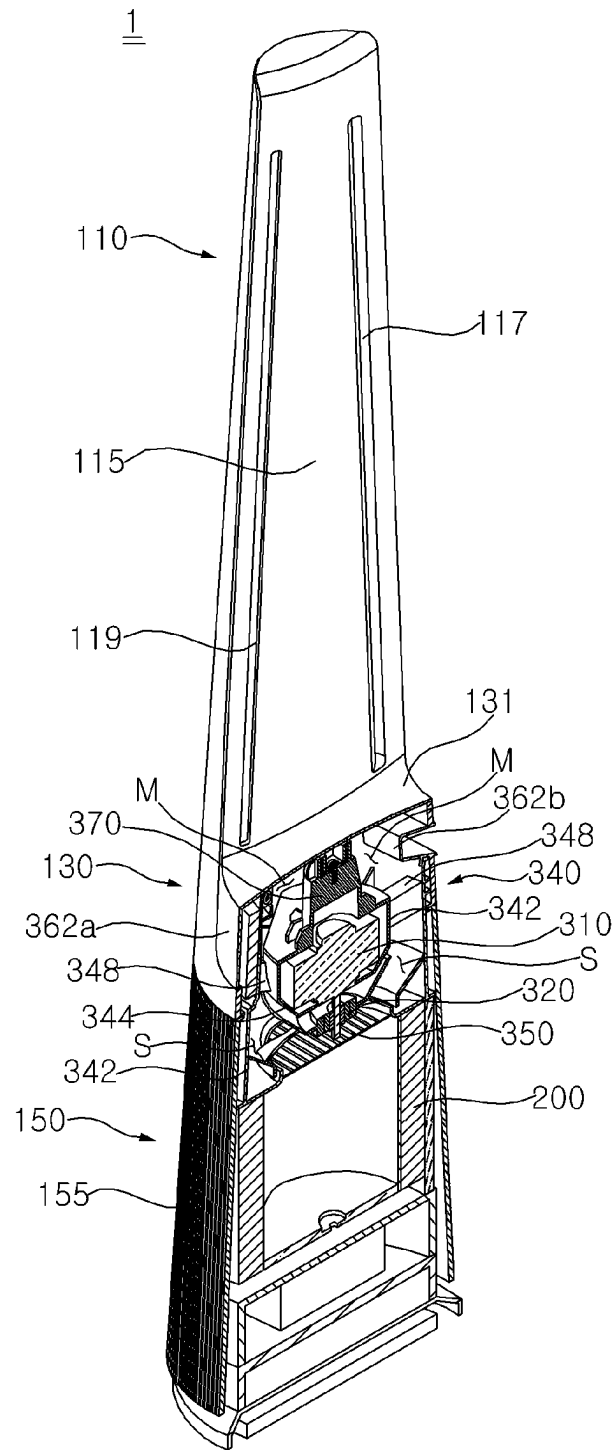


FIG. 8

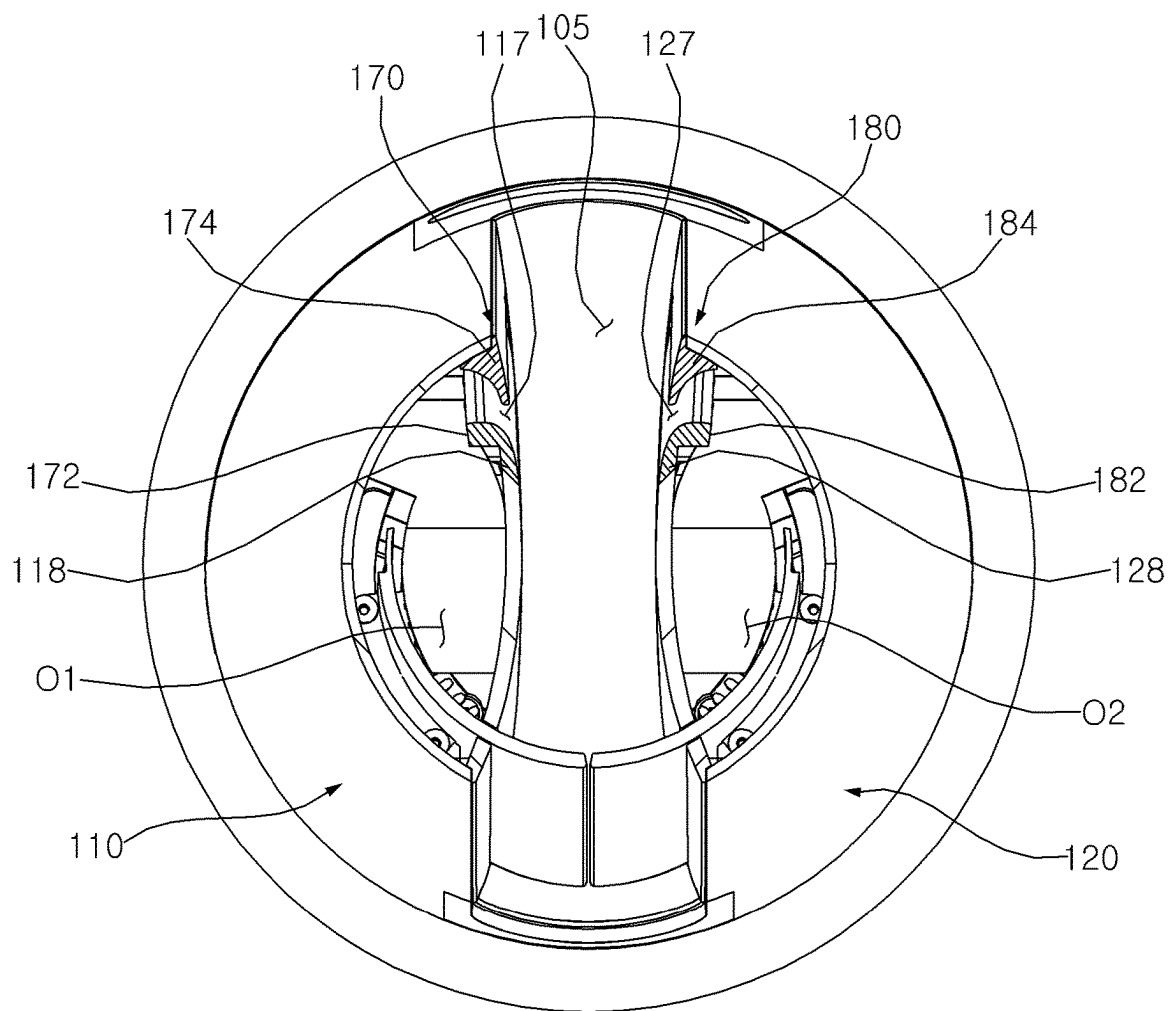


FIG. 9

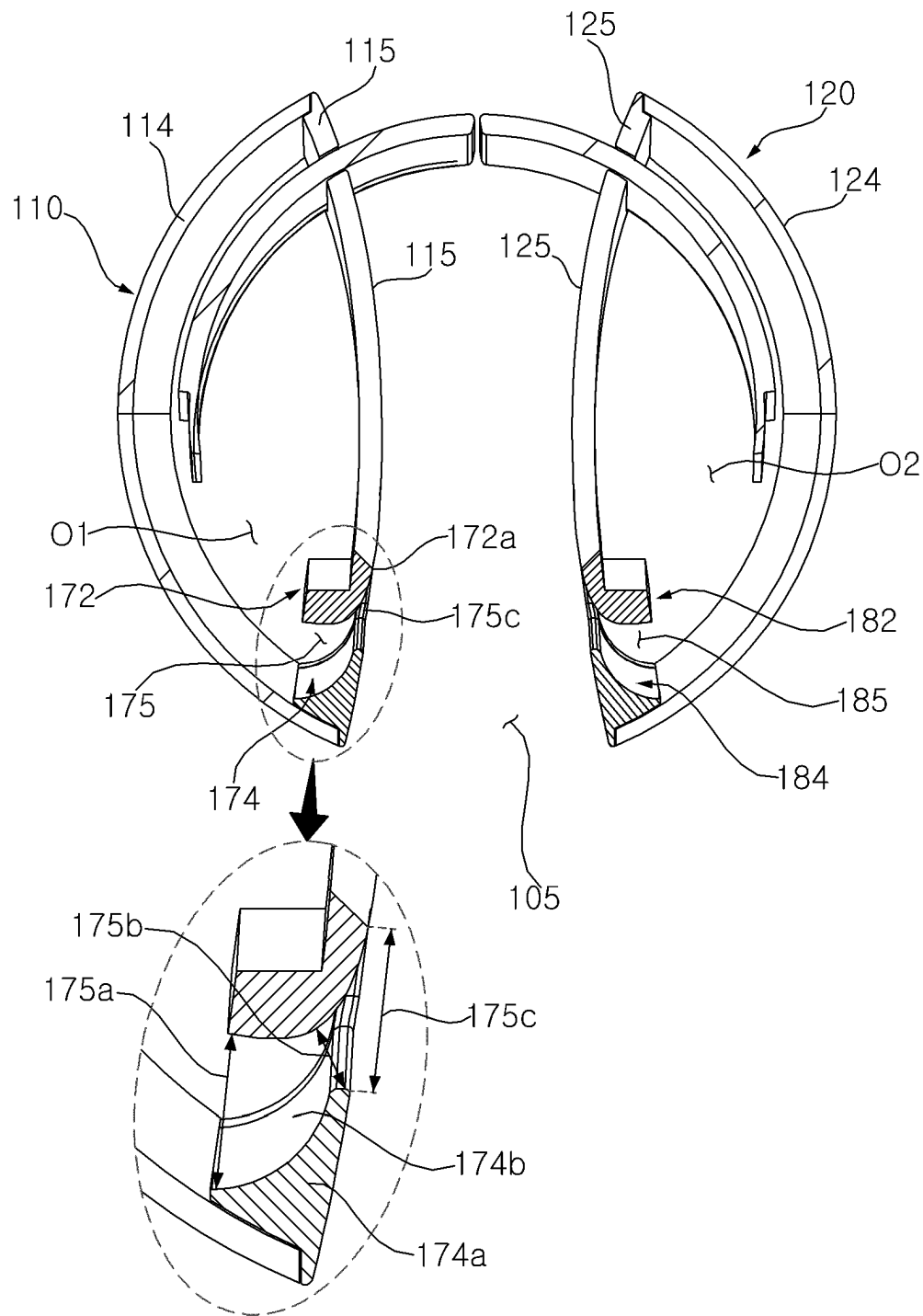


FIG. 10

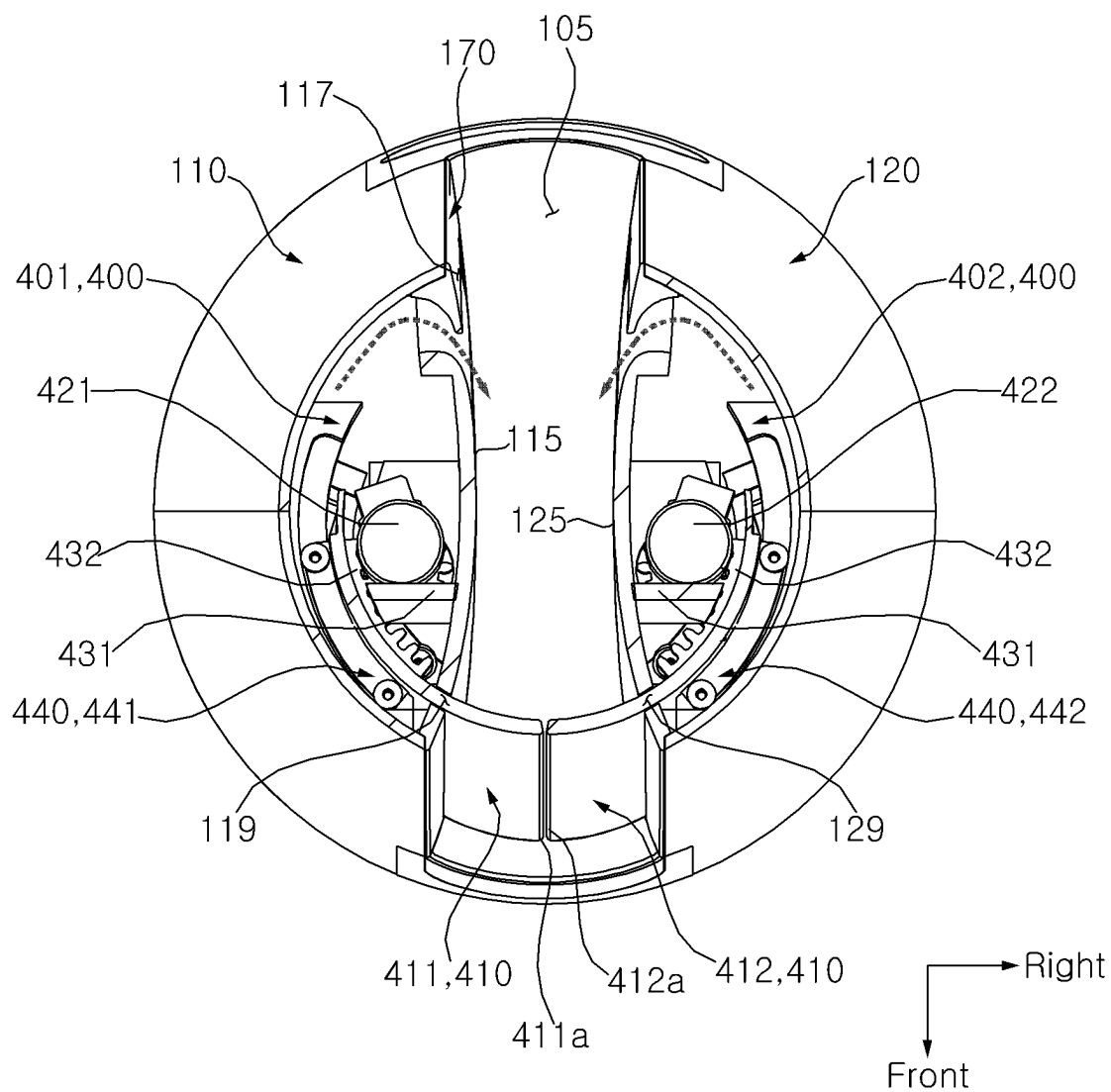


FIG. 11

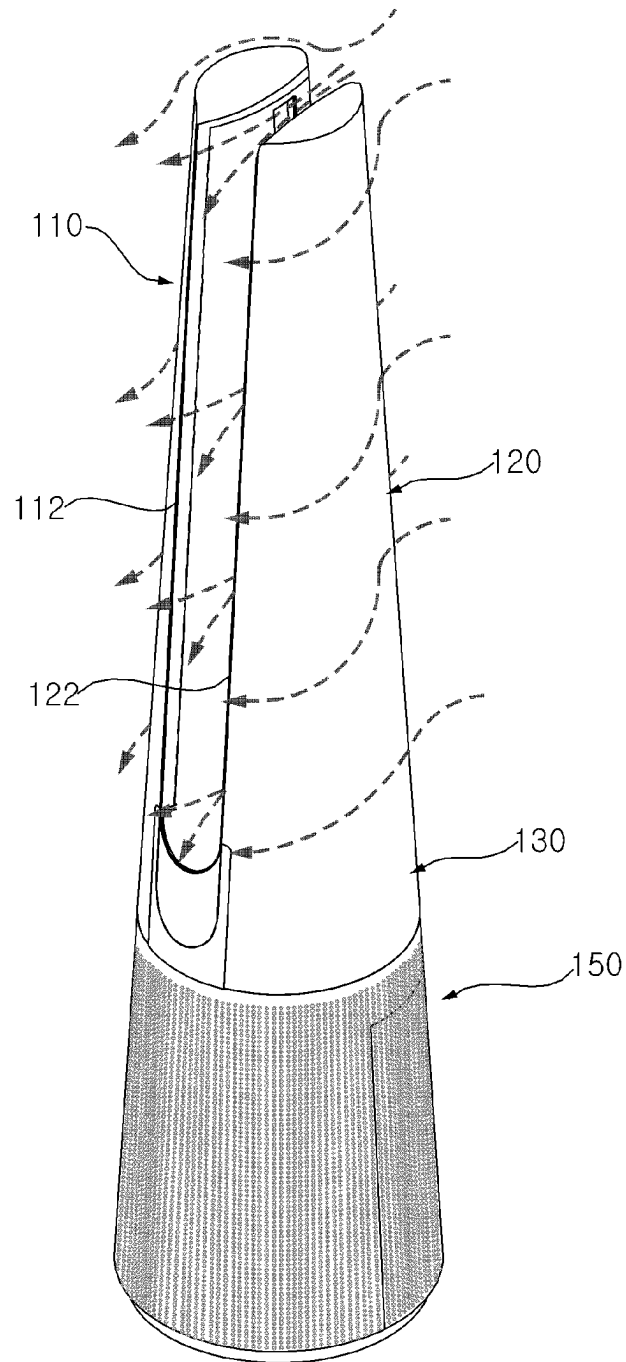


FIG. 12

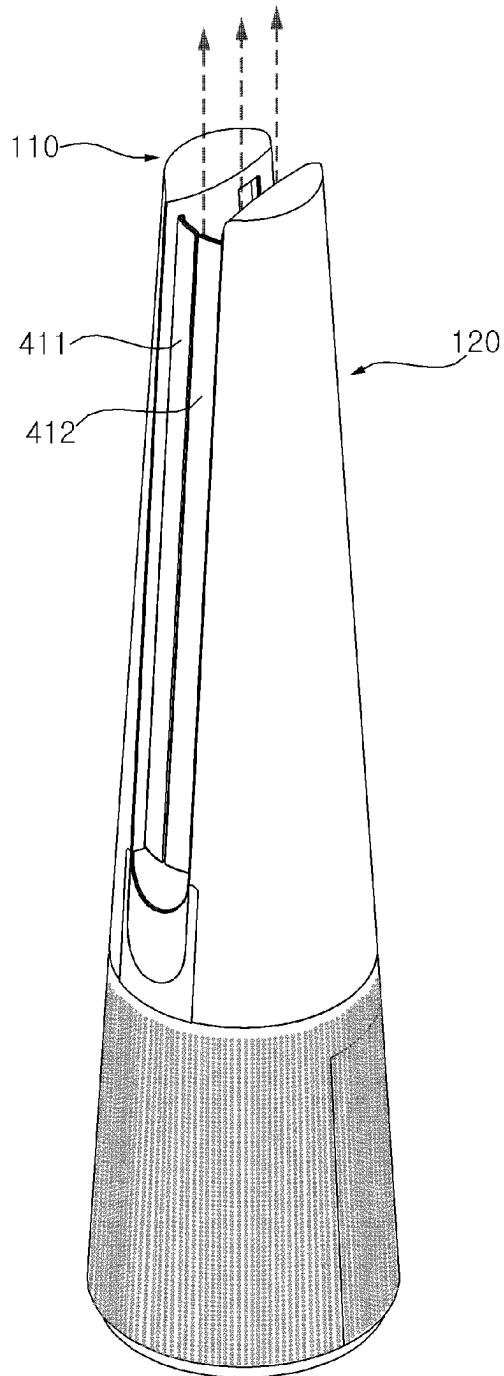


FIG. 13

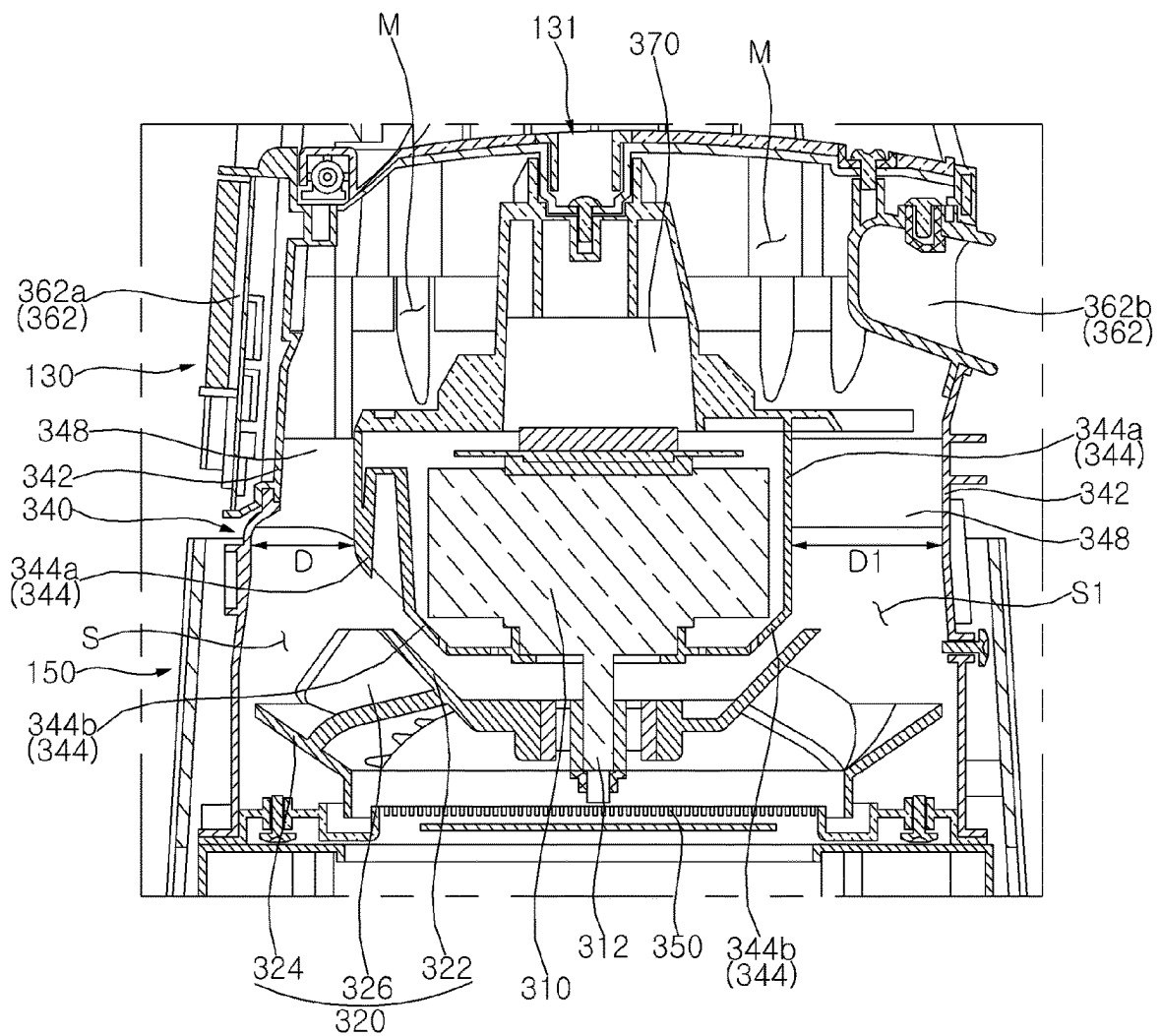


FIG. 14

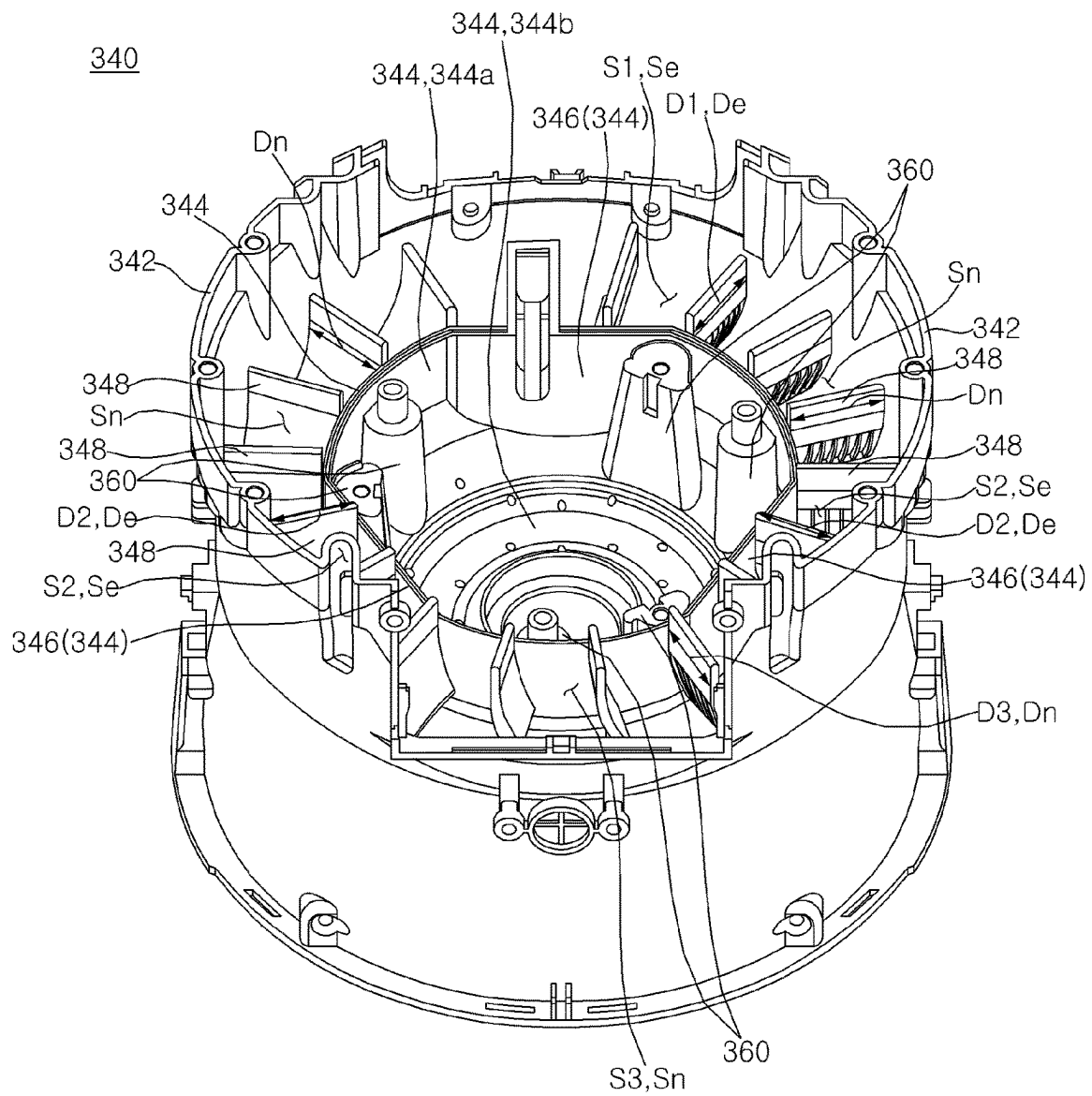


FIG. 15

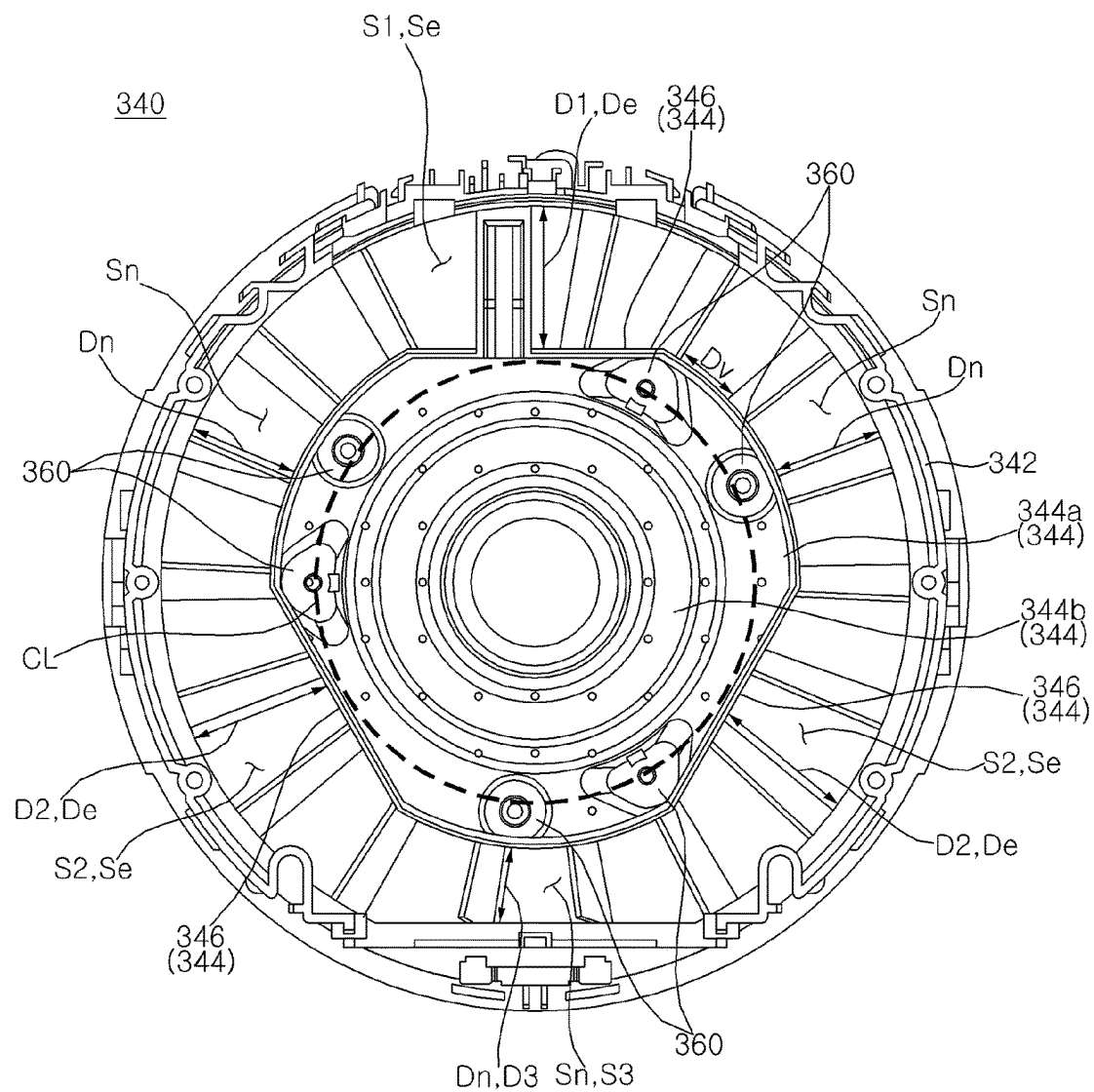


FIG. 16

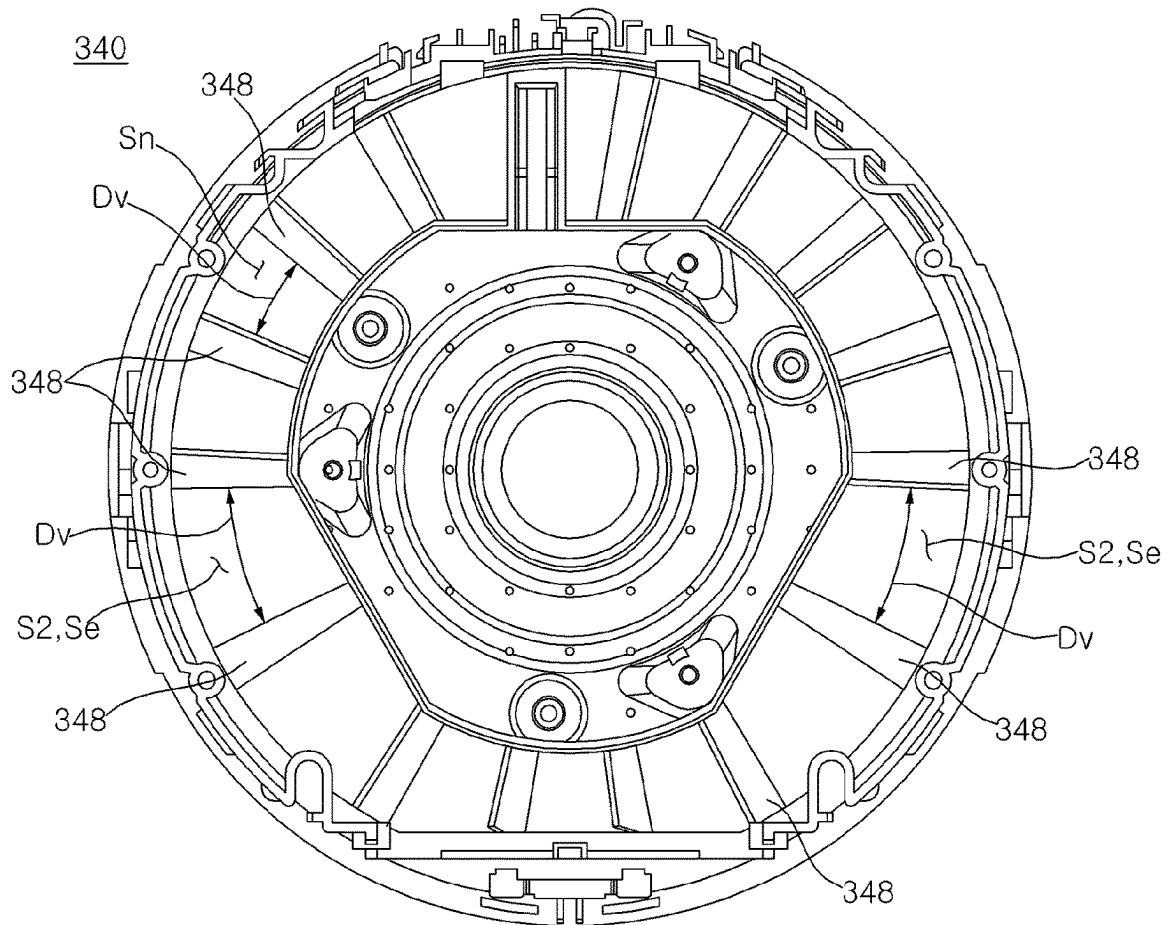
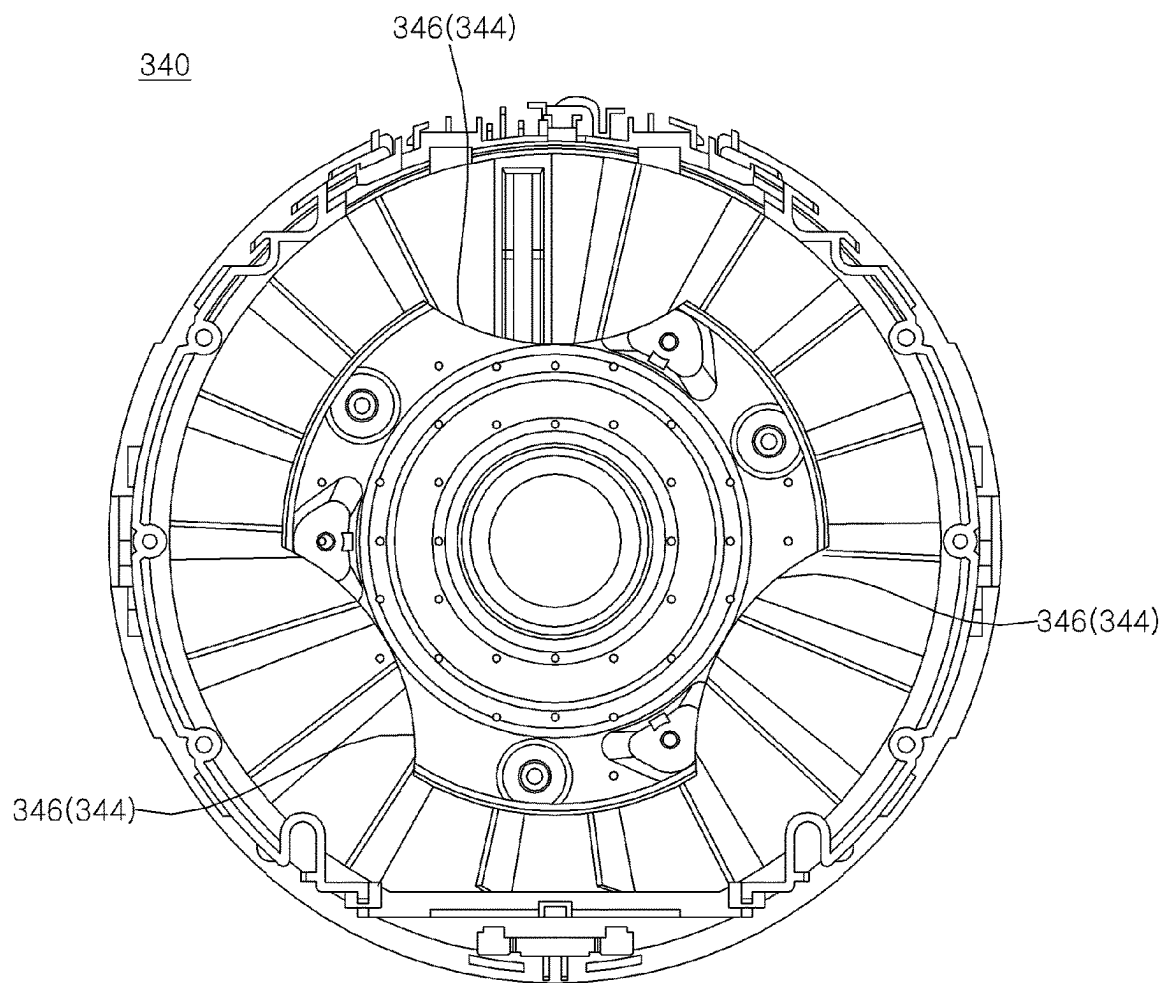


FIG. 17



1

BLOWER

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

This application claims priority to Korean Patent Application No. 10-2022-0146104, filed Nov. 4, 2022, whose entire disclosures are hereby incorporated by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present disclosure relates to a blower for discharging air using the Coanda effect, and more particularly to a blower including a diffuser for guiding a flow.

2. Description of the Related Art

A blower, having a suction port and a discharge port and including a blower fan therein, is a device for discharging air, drawn in through the suction port, to the outside through the discharge port. In this case, it is known that the blower should be provided with an appropriate interior space (or an air passage inside the blower), since the space is directly related to air-blowing performance of the blower itself as well as energy efficiency (degree of vibration and noise, power efficiency, etc.) of the blower.

Generally, the blower includes a diffuser located downstream of the blower fan so as to allow air, discharged from the blower fan, to smoothly flow to a discharge side. The diffuser is a device for increasing static pressure at the expense of a fluid speed, and for guiding a fluid in a desired direction through a vane which is a flow resistor designed with intention. That is, the diffuser in the blower is a device for allowing air, discharged from the blower fan, to pass through the vane to flow to the discharge side.

In an existing blower disclosed in Chinese patent publication Nos. 111156623 A and 107023884 A, if the aforementioned diffuser is located downstream of the fan, an air passage formed in the diffuser is uniformly manufactured, such that the air passage has a ring shape arranged in a concentric circle around a rotation axis of the fan.

However, in many cases, a position of an interior space of the blower may not be radially symmetric about the rotation axis of the fan. For example, an asymmetric structure may be disposed only on one side of the interior space of the blower, or if the discharge port of the blower may have a tower shape instead of the ring shape. As the diffuser is uniformly formed in the ring shape even when the interior space is asymmetric about the rotation axis, the existing blower has a low flow efficiency, thereby degrading the air blowing performance of the blower.

In addition, a diffuser, which is integrally injection molded, is generally used in the existing blower. In this case, even when there is a structural change by upgrading internal components of the blower or further installing new components such as a heater, lighting fixture, etc., the shape of the diffuser may not be appropriately adjusted according to the changed structure.

Further, the existing blower has a structure in that the components are generally stacked from bottom to top with the diffuser disposed therebetween, such that a load at the top of the diffuser is concentrated on the diffuser, but an outer circumference and an inner circumference of the existing blower are merely connected by a thin vane, without

2

a separate reinforcing structure for sharing the load, thereby degrading durability of the diffuser.

SUMMARY OF THE INVENTION

It is an objective of the present disclosure to solve the above and other problems.

It is another objective of the present disclosure to provide a blower with an improved air-blowing performance by providing a diffuser in a shape corresponding to a shape of a passage inside the blower.

It is yet another objective of the present disclosure to provide a blower capable of guiding a flow as well as safely supporting load even in a structure in which a load at the top is concentrated on the diffuser.

It is yet another objective of the present disclosure to provide a blower having excellent air-blowing performance by simply modifying an existing diffuser.

It is yet another objective of the present disclosure to provide a blower including a diffuser capable of being adjusted according to a structure when the structure is further installed later during use of the blower, such that excellent air-blowing performance of the blower may be maintained even when a passage inside the blower is changed.

The objectives of the present disclosure are not limited to the aforementioned objectives and other objectives not described herein will be clearly understood by those skilled in the art from the following description.

A blower according to an embodiment of the present disclosure includes: a case; a fan; and a diffuser disposed downstream of the fan and guiding air, discharged from the fan, toward the discharge ports.

The diffuser according to an embodiment of the present disclosure may include: an outer wall having a circumference; an inner wall spaced apart inwardly from the outer wall so that an air passage S, through which the air discharged from the fan flows, is formed therebetween.

The diffuser according to an embodiment of the present disclosure may include: a plurality of vanes extending radially between the outer wall and the inner wall and dividing the air passage S into a plurality of unit passages.

In this case, the outer wall and the inner wall may be formed to have areas with different radial separation distances D between the outer wall and the inner wall.

Accordingly, in an area of the air passage of the diffuser in which air is discharged to a portion having a high flow resistance in the blower, the radial separation distance D between the outer wall and the inner wall increases to enlarge the air passage S, thereby compensating for the asymmetry of the passage inside the blower, allowing a uniform volume of air to reach the entire area of the discharge ports, and improving the performance of the blower.

The case of the blower according to an embodiment of the present disclosure may include a lower case, a middle case, and/or an upper case which are sequentially stacked. The lower case may have the suction port formed therein, and the fan and the diffuser disposed therein. The upper case may have the discharge ports formed therein, may be disposed over the lower case, and may have a discharge passage O disposed therein through which the air flows toward the discharge ports. The middle case may be disposed between the lower case and the upper case, and may have an intermediate passage M to allow the air passage S to communicate with the discharge passage O.

The blower according to an embodiment of the present disclosure may further include a penetrating member having at least a portion penetrating into the middle case and disposed in the intermediate passage M. In this case, the separation distance D in a first area S1 may be larger than in an area adjacent to the first area S1 in the air passage S of the diffuser, the first area S1 being defined as a projected area in which the penetrating member is projected to the air passage S of the diffuser.

Accordingly, as the separation distance D in the first area S1 is larger than in the adjacent area, a volume of air per unit time passing through the first area S1 relatively increases, such that even with a local flow resistance caused by the penetrating member 362, a uniform volume of air may reach the entire area of the discharge ports, thereby improving the performance of the blower.

The upper case according to an embodiment of the present disclosure may include: a first tower and a second tower, each having a discharge passage O and the discharge port and communicating with the middle case, wherein the separation distance D in at least a portion of a second area S2 may be formed to be larger than in an area adjacent to the second area in the air passage S of the diffuser, the second area S2 being defined as an area in which the first discharge passage O1 and the second discharge passage O2 are projected to the air passage S.

By adjusting the discharge passages, connected to each of the plurality of discharge ports, according to a passage width of the diffuser, the air may flow directly and straight toward the discharge ports, thereby increasing the proportion of air reaching the discharge ports and reducing the flow resistance in the blower. Further, the volume of air may be distributed uniformly to each of the plurality of discharge ports, such that the performance of the blower may be improved.

In the blower according to an embodiment of the present disclosure, the first tower and the second tower are laterally spaced apart from each other, such that a space may be formed therebetween. In this case, the blower may further include: a bridge surface connecting lower ends of side walls of the first tower and the second tower which face each other with respect to the space; and a supporter having a lower portion connected to an upper end of the inner wall and an upper portion connected to a lower side of the bridge surface, so as to transfer load from the bridge surface to the inner wall.

In this case, in order to prevent the load at the top of the blower from being concentrated on the inner wall, the plurality of vanes may have radially inner ends connected to the inner wall and radially outer ends connected to the outer wall.

Accordingly, in a structure in which it is required to stably transfer the load at the top of the blower to the bottom of the blower, particularly in the blower having the shape of twin towers which are spaced apart from each other, the inner wall and the outer wall are connected by the vanes of the diffuser for guiding an air flow while transferring the load from the top of the blower 1 to the inner wall 344 of the diffuser, thereby distributing the burden of load of the inner wall to the outer wall, and improving structural rigidity of the diffuser.

In the diffuser of the blower according to an embodiment of the present disclosure, the air passage S may include an extended area Se in which the separation distance D is greater than in other adjacent areas, wherein a recessed portion, which is recessed radially inward, may be formed in the inner wall of the extension area Se.

Accordingly, compared to the case where the outer wall is bent outward to widen the passage of the diffuser, the inner wall is bent inward, thereby simply widening the passage of the diffuser while maintaining a total volume of the diffuser, such that an air volume of the blower may increase.

The diffuser of the blower according to an embodiment of the present disclosure may further include a sub-column disposed within the inner wall, disposed on a same circumference as the recessed portion with respect to a center of the inner wall, and having an upper end connected to the lower portion of the supporter.

Accordingly, the sub-column disposed within the inner wall and the recessed portion of the inner wall may support the load at the top of the blower on the same circumference, thereby preventing damage to the inner wall when the load is concentrated on the recessed portion of the inner wall, and improving structural stability of the blower.

According to an embodiment of the present disclosure, the plurality of vanes may be equally spaced apart around a circumference of the inner wall or a circumference of the outer wall. In addition, according to another embodiment of the present disclosure, the separation distance between adjacent vanes of the plurality of vanes may be larger in the extended area Se than in other adjacent areas.

While the vanes may serve to guide a flow in an upward direction, the vanes may also locally increase flow resistance. Thus, an air volume in a specific area may be improved by reducing the number of vanes in an area in which it is required to secure the air volume.

According to an embodiment of the present disclosure, a plurality of grooves may be formed circumferentially in an inner surface of the outer wall and an outer surface of the inner wall. Separation distances between adjacent grooves of the plurality of grooves may be equally spaced apart. The plurality of vanes may be press-fitted into predetermined grooves of the plurality of grooves, to be fastened thereto.

A separation distance between the vanes may be adjusted simply by employing a groove-fitting method and allowing the vanes to be selectively fastened to grooves at desired positions, among a plurality of predetermined grooves. For example, even when an internal structure of the blower is changed by a user's adding an option and the like, it is possible to simply adjust the blower according to the changed structure.

In the diffuser of the blower according to an embodiment of the present disclosure, a bent portion may be a curved surface that is convex inward. Accordingly, as the bent portion is formed as a curved surface that is convex inward, the width of the air passage S may further increase compared to the case where the bent portion is simply formed as a planar surface.

Other detailed matters of the exemplary embodiments are included in the detailed description and the drawings.

Effects of the Invention

According to the present disclosure, the air passage S is widened in a predetermined area of the diffuser, thereby compensating for the asymmetry (e.g., non-annular shape of the discharge port of the blower or the penetrating member disposed downstream of the diffuser) of the passage inside the blower, allowing a uniform volume of air to reach the entire area of the discharge ports, and improving blowing performance and noise performance of the blower.

According to the present disclosure, in a structure in which it is required to stably transfer the load at the top of the blower to the bottom of the blower (particularly in the

5

blower having the shape of twin towers which are spaced apart from each other), the inner wall and the outer wall are connected by the vanes of the diffuser for guiding an air flow while transferring the load from the top of the blower to the inner wall of the diffuser, thereby distributing the burden of load of the inner wall to the outer wall, and improving structural rigidity of the diffuser.

According to the present disclosure, compared to the case where the outer wall is bent outward to widen the passage of the diffuser, the inner wall is bent inward, thereby simply widening the passage of the diffuser while maintaining a total volume of the diffuser, such that an air volume of the blower may increase.

According to the present disclosure, the sub-column disposed within the inner wall and the recessed portion of the inner wall may support the load at the top of the blower on the same circumference, thereby preventing damage to the inner wall when the load is concentrated on the recessed portion of the inner wall, and improving structural stability of the blower.

According to the present disclosure, an air volume in a specific area may be improved by reducing the number of vanes in an area in which it is required to secure the air volume.

According to the present disclosure, a separation distance between the vanes may be adjusted simply by employing a groove-fitting method and allowing the vanes to be selectively fastened to grooves at desired positions, among a plurality of predetermined grooves. Accordingly, even when an internal structure of the blower is changed by a user's adding an option and the like, it is possible to simply adjust the blower according to the changed structure.

According to the present disclosure, as the bent portion is formed as a curved surface that is convex inward, the width of the air passage S may further increase compared to the case where the bent portion is simply formed as a planar surface.

The effects of the present disclosure are not limited to the aforesaid, and other effects not described herein will be clearly understood by those skilled in the art from the following description of the appended claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a blower according to an embodiment of the present disclosure.

FIG. 2 is an exemplary view illustrating an example of an operation of FIG. 1.

FIG. 3 is a front view of FIG. 2.

FIG. 4 is a plan view of FIG. 3.

FIG. 5 is a right cross-sectional view of FIG. 2.

FIG. 6 is a front cross-sectional view of FIG. 2.

FIG. 7 is a cross-sectional perspective view of FIG. 1.

FIG. 8 is a plan view taken along line XI-XI of FIG. 3.

FIG. 9 is a bottom view taken along line IX-IX of FIG. 3.

FIG. 10 is a plan view taken along line IX-IX of FIG. 3.

FIG. 11 is an exemplary view illustrating a horizontal airflow of a blower according to an embodiment of the present disclosure.

FIG. 12 is an exemplary view illustrating an ascending airflow of a blower according to an embodiment of the present disclosure.

FIG. 13 is an enlarged view of portion TA of FIG. 5.

FIG. 14 is a perspective view of a diffuser according to an embodiment of the present disclosure.

FIG. 15 is a plan view of FIG. 15

6

FIG. 16 is a plan view of a diffuser according to another embodiment of the present disclosure.

FIG. 17 is a plan view of a diffuser according to yet another embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Advantages and features of the present disclosure and methods of accomplishing the same will be more clearly understood from exemplary embodiments described below with reference to the accompanying drawings. However, the present disclosure is not limited to the following embodiments but may be implemented in various different forms.

The embodiments are provided only to complete disclosure of the present disclosure and to fully provide a person having ordinary skill in the art to which the present disclosure pertains with the category of the present disclosure, and the present disclosure is only defined by the scope of the appended claims. Wherever possible, like reference numerals generally denote like elements through the specification.

Hereinafter, unless otherwise specified, the terms "upstream" and "downstream" may refer to upstream and downstream sides of an airflow in a blower 1. The following description may be based on the premise that a suction port 155, a fan 320, a diffuser 340, and discharge ports 117 and 127 are arranged from bottom to top, unless otherwise specified. However, this is merely for convenience of explanation, and the gist of the present disclosure should be understood based on whether an airflow direction is an upstream direction or a downstream direction, and the technical idea of the present disclosure should not be construed as being limited by left, right, up, and down directions thereof. For example, if the suction port 155 is disposed above the diffuser 340, and the discharge ports 117 and 127 are disposed below the diffuser 340, a passage width of the diffuser 340 may be formed in consideration of a lower space of the diffuser 340.

The following description will be given based on the premise that an inner wall 344 of the diffuser 340 functions as a motor housing that accommodates a fan motor 310. However, the technical idea of the present disclosure is not limited thereto, and it should be understood that the gist of the present disclosure lies in technical features related to the airflow of the diffuser 340. For example, the fan motor 310 may be coupled to an upstream end of the fan 320, and the diffuser 340 may be disposed at a downstream end of the fan 320.

A blower 1 according to an embodiment of the present disclosure includes: a case 100; a fan 320; and a diffuser 340 guiding air, wherein the diffuser 340 includes: an inner wall 344 and an outer wall 342 spaced apart from each other and each having a circumference, with an air passage S formed therebetween; and a plurality of vanes 348 extending radially between the outer wall 342 and the inner wall 344 and dividing the air passage S into a plurality of unit passages ss, wherein the outer wall 342 and the inner wall 344 are formed to have areas with different radial separation distances D between the outer wall 342 and the inner wall 344.

Hereinafter, the overall configuration of the blower 1 and the diffuser 340 according to an embodiment of the present disclosure will be described with reference to FIGS. 1, 5 to 7, and 14 to 19. First, the overall configuration of the blower 1 will be described below.

The case 100 may form the exterior of the blower 1. The case 100 has a suction port 155 and discharge ports 117 and 127.

The case **100** may include a lower case, a middle case **130**, and an upper case **140**.

The suction port **155** may be formed in the lower case. The suction port **155** may be open at the circumference of the lower case.

The fan **320** and the diffuser **340** may be disposed in the lower case. With respect to an air flow, the suction port **155**, the fan **320**, and the diffuser **340** may be sequentially disposed in the lower case. The diffuser **340** may serve to reduce a radial component and reinforce an upward component in the air flow. The diffuser **340** may be disposed at a lowermost downstream end of the lower case. The lowermost downstream end of the diffuser **340** may be a downstream end of the lower case (see FIGS. **5** to **7**).

The discharge ports **117** and **127** may be formed in the upper case **140**. The discharge ports **117** and **127** may be open at the circumference of the upper case **140**. Alternatively, as will be described later, the discharge ports **117** and **127** may be formed in the open portion of the upper case **140**, such that the discharge ports **117** and **127** may be formed in a unique shape.

The upper case **140** may be disposed over the lower case. A discharge passage **O**, through which air flows toward the discharge ports **117** and **127**, may be formed in the upper case **140**. The discharge passage **O** may be understood as an inner space of the upper case **140**.

The middle case **130** may be disposed between the lower case and the upper case **140**. The middle case **130** may include an intermediate passage **M** allowing the air passage **S**, which will be described later, to communicate with the discharge passage **O**. The intermediate passage **M** may be understood as an inner space of the middle case **130**.

A downstream end of the lower case may be a downstream end of the air passage **S** of the diffuser **340**. An upstream end of the intermediate passage **M** may communicate with the air passage **S**. A downstream end of the intermediate passage **M** may communicate with the discharge passage **O**. The air passage **S**, the intermediate passage **M**, and the discharge passage **O** may form a continuous space.

A lower end of the middle case **130** may be connected to an upper end of the lower case. An upper end of the middle case **130** may be connected to a lower end of the upper case **140**. The lower case, the middle case **130**, and the upper case **140** may form a continuous surface.

Air discharged from the diffuser **340** may pass through the intermediate passage **M** to be supplied into the discharge passage **O**.

Hereafter, the middle case **130** may be used interchangeably with and/or referred to as a "tower base **130**." The tower base **130** may connect lower portions of the first tower **110** and the second tower **120**.

The fan **320** may be disposed in the case **100**. The fan **320** may draw in outside air through the suction port **155**, and then discharges the air toward the discharge ports **117** and **127**. Any type of fan may be used depending on an internal structure of the blower **1**. It is preferred to use a mixed flow fan **320** to reduce vibrations and noise in the blower **1**.

Hereinafter, the diffuser will be described particularly with reference to FIGS. **5** to **7** and **13** to **19**.

The diffuser **340** is disposed downstream of the fan **320** and guides the air, discharged from the fan **320**, toward the discharge ports **117** and **127**.

The diffuser **340** includes: an outer wall **342** having a circumference; an inner wall **344** spaced apart inwardly from the outer wall **342** so that the air passage **S**, through which the air discharged from the fan **320** flows, is formed

therebetween; and a plurality of vanes **348** extending radially between the outer wall **342** and the inner wall **344** and dividing the air passage **S** into a plurality of unit passages **ss**.

The outer wall **342** may have a circumference with both ends that are open. The inner wall **344** may have a circumference that is narrower in width than the outer wall **342** and may be disposed within the outer wall **342**. The inner wall **344** may face the outer wall **342**. The air passage **S** may be formed in a space between the inner wall **344** and the outer wall **342** that face each other. The inner wall **344** and the outer wall **342** may be substantially parallel to each other.

The outer wall **342** may be elongated toward an upstream side in an air flow direction and may accommodate the fan **320** therein. The inner wall **344** is shorter than the outer wall **342**, and the fan **320** may be disposed on an upstream side of the inner wall **344**. Each of the fan **320** and the inner wall **344** may face the outer wall **342**.

The inner wall **344** may accommodate a fan motor **310** therein which provides a driving force to the fan **320**. The fan motor **310** accommodated in the inner wall **344** may be coupled to the fan **320** via a motor shaft **312** that passes through one side of the inner wall **344**. The motor shaft **312** of the fan motor **310** may be coupled to a rotating shaft of the fan **320**. As the fan motor **310** is accommodated in the inner wall **344**, space efficiency of the blower **1** may be improved. In order to accommodate the fan motor **310** therein and to support a lower surface of the fan motor **310** from below, the inner wall **344** may further include a sub-part **344b** that covers a lower part of the fan motor **310** (see FIG. **13**). The sub-part **344b** may be formed in a cone shape having a cross-section that decreases toward the fan **320**. The sub-part **344b** may have an open lower portion, through which the motor shaft **312** of the fan motor **310** may pass. The motor shaft **312**, having passed through the open portion of the sub-part **344b**, may be coupled to the rotating shaft of the fan **320**. In order to distinguish from the sub-part **344b** of the inner wall **344**, a portion of the inner wall **344** that faces the outer wall **342** may be referred to as a main part **344a**.

The sub-part **344b** may extend radially inward from a lower end of the main part **344a** and obliquely toward the fan **320**. At a portion where the fan motor **310** and the fan **320** are close enough to be connected to each other by the motor shaft **312**, the sub-part **344b** may extend radially inwardly and horizontally. A central portion of the horizontally extending portion may be open such that the motor shaft **312** may pass therethrough.

However, the technical idea of the present disclosure is not limited thereto, and a motor housing accommodating the fan motor **310** and the inner wall **344** may also be provided separately.

An inlet (or upstream end) of the air passage **S** of the diffuser **340** may be connected to a discharge portion of the fan **320**, thereby allowing the air discharged from the fan **320** to be supplied to the diffuser **340**. An outlet (or downstream end) of the air passage **S** of the diffuser **340** may be connected to the intermediate passage **M** of the tower base **130**, thereby allowing the air discharged from the diffuser **340** to be supplied to the tower base **130**.

The air passage **S** may be understood as a space between the outer wall **342** and the inner wall **344**. The air passage **S** may have a substantially ring shape. The air discharged from the fan **320** may flow through the air passage **S** to be discharged from the diffuser **340**. The air passage **S** may be simply abbreviated to a "passage of the diffuser **340**" or a "passage formed by the diffuser **340**."

In this specification, a radial direction refers to not only a linear direction extending from the center of a circle to the circumference thereof, but also a linear direction extending from the center to a periphery of any shape as long as the center and periphery of the shape may be defined. That is, the outer wall **342** and/or the inner wall **344** are not necessarily formed in a circular shape.

The air passage **S** may be divided into a plurality of unit passages **ss** by the plurality of vanes **348**. The air passage **S** may be understood as a sum of the plurality of unit passages **ss**. The vanes **348** may be elongated radially to be disposed between the outer wall **342** and the inner wall **344**. The plurality of vanes **348** may be spaced apart from each other to be arranged along the circumference of the inner wall **344** or the circumference of the outer wall **342**. The vanes **348** of the diffuser **340** may guide the air discharged from the blower fan **320** in an upward direction, and may prevent the air, having passed through the diffuser **340**, from turning (i.e., backflow) toward the blower fan **320**.

The outer wall **342** and the inner wall **344**, which are spaced apart from each other, may define a radial separation distance **D** therebetween. The separation distance **D** at a specific portion may be equal to a radial length of the vane **348** disposed at the corresponding portion. In a general blower diffuser **340**, each of the outer wall **342** and the inner wall **344** is formed in a cylindrical shape, and thus the separation distance **D** is the same for all the radii of rotation.

However, according to an embodiment of the present disclosure, the outer wall **342** and the inner wall **344** are formed to have areas with different radial separation distances **D** between the outer wall **342** and the inner wall **344**.

By providing different separation distances **D** of the diffuser **340** in a predetermined area, parameters, such as air volume, pressure, etc., of air passing through the diffuser **340** may be adjusted differentially. For example, as illustrated in FIG. **13**, the air passage **S** may include a first area **S1** with a first separation distance **D1** which is a relatively long separation distance **D**, and another area (left space of the inner wall **344** in FIG. **13**) with the separation distance **D** which is shorter than the first separation distance **D1**.

Accordingly, for example, in an area of the air passage of the diffuser **340** in which air is discharged to a portion having a high flow resistance in the blower **1**, the radial separation distance **D** between the outer wall **342** and the inner wall **344** increases to widen the air passage **S**, thereby compensating for the asymmetry of the passage inside the blower, allowing a uniform volume of air to reach the entire area of the discharge ports **117** and **127**, and improving the performance of the blower **1**.

A portion of the outer wall **342** and/or the inner wall **344** may be bent radially outward or inward, so that the outer wall **342** and/or the inner wall **344** may have areas with different radial separation distances **D** between the outer wall **342** and the inner wall **344**.

The outer wall **342** may form the exterior of the diffuser **340**, in which the diffuser **340** is required to be accommodated in a limited internal space of the blower **1**, such that the outer wall **342** may be formed in a shape that is in inner contact with the case **100** of the blower **1**. Accordingly, it may be preferred that for adjusting the separation distance **D**, it may be more efficient to bend the inner wall **344** in a radially inward direction than in a radially outward direction. A method of adjusting the separation distance **D** will be described in detail later.

Hereinafter, the “separation distance **D**” may directly refer to a radial distance between the outer wall **342** and the inner wall **344**. The separation distance **D** may be under-

stood as a width of the passage formed by the diffuser **340**. Hereinafter, the “width of the passage of the diffuser **340**” may be used interchangeably with the “separation distance **D** between the outer wall **342** and the inner wall **344**” to indicate the same meaning.

The blower **1** may further include a penetrating member **362** having at least a portion penetrating into the middle case **130** to be disposed on the intermediate passage **M** (see FIGS. **5** to **7** and **13**).

The penetrating member **362** may be a structure for providing a specific function to the blower **1**. The penetrating member **362** may occupy a portion of the intermediate passage **M**. The flow of air in the intermediate passage **M** may be disturbed by the penetrating member **362**. The penetrating member **362** may be a resistance to the flow in the intermediate passage **M**. In order to compensate for the flow resistance in the intermediate passage **M**, the separation distance **D** of the passage width of the diffuser **340** may be adjusted as will be described later.

For example, the penetrating member **362** may be a handle **362b** that facilitates movement of the blower **1** by allowing a user to put their hands into a recessed portion. The handle **362b** may be recessed inward from an outer surface of the middle case **130**, and a side opposite a direction of the recessed portion may be open to communicate with an external space.

Alternatively, the penetrating member **362** may be, for example, a display unit **362a** including a display for visually displaying information about the blower **1** to a user, and a display-related component. The display may be disposed on an outer surface of the middle case **130** to be exposed to the outside. The display-related component may be a display electronic circuit board (PCR) that penetrates from the display into the middle case **130** and is electrically connected to the display.

There may be a plurality of penetrating members **362**. For example, the display unit **362a** may be disposed on a front surface of the middle case **130**, and the handle **362b** may be disposed on a rear surface of the middle case **130**.

The first area **S1** in the air passage **S** of the diffuser **340** is defined as a projected area in which the penetrating member **362** is projected to the air passage **S** of the diffuser **340**, and the separation distance **D** may be formed to be larger in the first area **S1** than in an area adjacent to the first area **S1** (see FIGS. **13** to **15**).

In this case, based on the premise of straightness of air flow indicating that air discharged from the blower fan **320** flows straight, the projected area in which a specific component is projected to the air passage **S** of the diffuser **340** may refer to a specific area, from which air to flow toward a specific component is discharged, in the air passage **S** of the diffuser **340**. Hereinafter, the “area in which a specific component is projected to the air passage **S**” may be simply referred to as a “projected surface” or a “projected area.”

The first area **S1** may refer to a specific area, from which air to flow toward the penetrating member **362** is discharged, in the air passage **S** of the diffuser **340**. The first area **S1** may be understood as a virtual area in which the penetrating member **362** is vertically projected to the air passage **S** of the diffuser **340**. The air discharged from the projected area in the air passage of the diffuser **340** may flow toward the penetrating member **362**. The separation distance **D** in the first area **S1** may be referred to as the first separation distance **D1**.

As the separation distance **D** in the first area **S1** is larger than that in an adjacent area, a volume of air passing through the first area **S1** and discharged from the diffuser **340** may

11

be greater than a volume of air passing through the adjacent area and discharged from the diffuser 340. Accordingly, even with a local flow resistance caused by the penetrating member 362, a volume of air per unit time that reaches the discharge ports 117 and 127 after passing through the first area S1 may be similar to a volume of air that reaches the discharge ports 117 and 127 after passing through the adjacent area.

The upper case 140 may include a first discharge case 110 and a second discharge case 120 (see FIG. 1).

The first discharge case 110 may have the first discharge port 117. The first discharge case 110 may include a first discharge passage O1 formed therein, through which air flows toward the first discharge port 117. The first discharge port 117 may be formed as an opening in the first discharge case 110. The first discharge passage O1 may be understood as an inner space of the first discharge case 110. The air flowing through the first discharge passage O1 may be discharged through the first discharge port 117. The first discharge case 110 may have a cylindrical shape or a truncated cone shape.

The first discharge case 110 may be referred to as the first tower 110. The first tower 110 may have a tower shape that is elongated upward. The first discharge port 117 may also be elongated in a longitudinal direction of the first tower 110. However, the technical idea of the present disclosure is not limited by the shape of the first discharge case 110, and may be applied regardless of the shape of the first discharge case 110.

The second discharge case 120 may have the second discharge port 127. The second discharge case 120 may include a second discharge passage O2 formed therein, through which air flows toward the second discharge port 127. The second discharge port 127 may be formed as an opening in the second discharge case 120. The second discharge passage O2 may be understood as an inner space of the second discharge case 120. The air flowing through the second discharge passage O2 may be discharged through the second discharge port 127. The second discharge case 120 may have a cylindrical shape or a truncated cone shape.

The second discharge case 120 may be referred to as the second tower 120. The second tower 120 may have a tower shape that is elongated upward. The second tower 120 is spaced apart laterally from the first tower 110 such that a space BS may be formed therebetween. However, the technical idea of the present disclosure is not limited by the shape of the second discharge case 120, and may be applied regardless of the shape of the second discharge case 120.

The second discharge case 120 is spaced apart from the first discharge case 110, such that the space BS may be formed therebetween. The space BS may be a space in which the air discharged from the first discharge port 117 and the second discharge port 127 flows, and may be referred to as a "blowing space 105." The first discharge case 110 and the second discharge case 120 may be symmetrical to each other with respect to the space BS formed therebetween.

Each of the first discharge passage O1 and the second discharge passage O2 may communicate with the intermediate passage M of the middle case 130. Specifically, the first discharge passage and the second discharge passage may be connected in series to the intermediate passage 130 and may be parallel to each other. The air discharged from the diffuser 340 may pass through the intermediate passage M to be distributed to each of the first discharge passage O1 and the second discharge passage O2 (see FIG. 6).

A second area S2 in the air passage S of the diffuser 340 is defined as an area in which the first discharge passage O1

12

and the second discharge passage O2 are projected to the air passage S, and the separation distance D in at least a portion of the second area S2 may be formed to be larger than in an area adjacent to the second area S2. The distribution of the separation distance D may be symmetrical with respect to a third area S3 in which the space BS is projected to the air passage S (see FIGS. 14 and 15).

In the air passage S of the diffuser 340, the second area S2 may refer to a specific area, from which some of the air, discharged from the diffuser 340, is discharged to flow straight (or almost straight) to the first discharge passage or the second discharge passage. The second area S2 may be understood as a virtual area in which the first discharge passage and the second discharge passage are vertically projected to the air passage S of the diffuser 340. The air discharged from the second area S2 may pass through the intermediate passage M without any resistance because of straightness, to flow directly to the first discharge passage or the second discharge passage. The separation distance D in the second area S2 may be referred to as a second separation distance D2. The separation distance D in the third area S3 may be referred to as a third separation distance D3. The third separation distance D3 may be smaller than the second separation distance D2.

The air discharged from an area adjacent to the second area S2 may be subjected to the flow resistance until the air reaches the discharge passage. Particularly, after air is discharged from the third area S3, which is one of areas adjacent to the second area S2, and flows almost straight through the intermediate passage M, the air first collides with a bridge surface 131 which will be described later, and then is changed in flow direction to be distributed to the first discharge passage O1 and the second discharge passage O2.

Accordingly, a passage width of the second area S2 is larger than that of the adjacent area, such that the volume of air, discharged from the second area S2, has a larger proportion of the total volume of air per unit time discharged from the diffuser 340, thereby minimizing internal resistance on a path of the air flow to reach the discharge ports 117 and 127.

As described above, by adjusting the discharge passages, connected to each of the plurality of discharge ports 117 and 127, according to a passage width of the diffuser 340, the air may flow directly and straight toward the discharge ports 117 and 127, thereby increasing the proportion of air reaching the discharge ports 117 and 127 and reducing the flow resistance in the blower 1. Further, the volume of air may be distributed uniformly to each of the plurality of discharge ports 117 and 127, such that the performance of the blower 1 may be improved.

In a structure of the blower 1 in which the lower case, the middle case 130, and the upper case 140 are stacked from bottom to top, it is required to provide a structure for supporting the load at the top of the blower 1. Particularly, as the first tower 110 and the second tower 120 are laterally spaced apart from each other, resistance of the blower 1 to external force to widen or narrow the two towers may be weak.

Accordingly, the bridge surface 131 may connect lower ends of side walls 115 and 125 of the first tower 110 and the second tower 120 that face each other with respect to the space BS (see FIGS. 1 and 5 to 7). The bridge surface 131 may be an upper surface of the tower base 130. The bridge surface 131 may form a continuous surface with an inner wall (first inner wall) 115 of the first tower 110. The bridge surface 131 may form a continuous surface with an inner wall (second inner wall) 125 of the second tower. The bridge

13

surface **131** may fix the first tower **110** and the second tower **120** and may receive load from the first tower **110** and the second tower **120**.

Some of the air, discharged from the diffuser **340** and flowing through the intermediate passage **M**, may reach the bridge surface **131** to be distributed to the first tower **110** and the second tower **120** by flowing along the bridge surface **131**. The bridge surface **131** may be formed as a curved surface that is convex downward, thereby facilitating distribution of the flow.

A supporter may have a lower portion connected to an upper end of the inner wall **344** and an upper portion connected to the lower part of the bridge surface **131**, such that a load may be transferred from the bridge surface **131** to the inner wall **344**. The supporter, disposed between the diffuser **340** and the bridge surface **131**, may connect the diffuser **340** and the bridge surface **131** (see FIGS. **5** to **7** and **13**).

Connection of the lower portion of the supporter to the upper end of the inner wall **344** may include not only a direct connection thereof, but also an indirect connection thereof by providing another structure therebetween.

The supporter may have a predetermined pillar shape in order not to interrupt the flow in the blower **1** while transferring the load. For example, in order to uniformly transfer the load to the inner wall **344**, a planar cross-section of a lower end of the supporter may correspond to a planar cross-section of the upper end of the inner wall **344**. In addition, the supporter may have a cone shape or a truncated cone shape in which the planar cross-section decreases upwardly, so as to reduce the flow resistance in the blower **1**.

As the entire load of the inner wall of the first tower **110** and the second tower **120** is transferred to the inner wall **344** via the supporter, there may be a problem with durability of the inner wall **344**. Accordingly, the plurality of vanes **348** may have radial inner ends which are connected to the inner wall **344**, and radial outer ends connected to the outer wall **342**. That is, the plurality of vanes **348** not only divide the air passage **S** but also serve to connect the inner wall **344** and the outer wall **342**, thereby distributing the load, applied to the inner wall **344**, to the outer wall **342**.

Accordingly, in a structure in which it is required to stably transfer the load from the top of the blower **1** to the bottom of the blower **1**, particularly in the blower **1** having the shape of twin towers which are spaced apart from each other, the inner wall **344** and the outer wall **342** are connected by the vanes **348** of the diffuser **340** for guiding an air flow while transferring the load from the top of the blower **1** to the inner wall **344** of the diffuser **340**, thereby distributing the burden of load of the inner wall **344** to the outer wall **342**, and improving structural rigidity of the diffuser **340**.

The air passage **S** of the diffuser **340** may include a normal area **Sn**, in which the separation distance **D** is a predetermined value (or a normal separation distance **Dn**), and an extended area **Se** in which the separation distance **D** is greater than the predetermined value **Dn** (see FIGS. **14** and **15**).

The normal area **Sn** may refer to an area in which the separation distance **D** is not extended compared to other areas in the air passage **S**. The normal area **Sn** may be a relative concept that is determined in comparison to other areas. A normal separation distance **Dn** may refer to the separation distance **D** of the normal area **Sn**. The normal area **Sn** may be a concept including the aforementioned third area **S3**. The normal separation distance **Dn** may be a concept including the aforementioned third separation distance **D3**.

14

The separation distance **D** between the inner wall **344** and the outer wall **342** of an existing diffuser **340**, in which the separation distance **D** is not separately extended or reduced, is generally adjusted according to a width of a discharge portion of the fan **320**. This is for allowing air discharged from the fan **320** to completely pass through the diffuser **340**. Accordingly, the normal separation distance **Dn** may be the same as or similar to the width of the discharge portion of the fan **320**. However, if necessary, the normal separation distance **Dn** may be smaller than the width of the discharge portion of the fan **320**, thereby increasing velocity or pressure of the air discharged from the blower **1**.

The extended area **Se** may refer to an area in which the separation distance **D** is extended compared to other areas in the air passage **S**. The extended area **Se** may be a relative concept that is determined in comparison to other areas. An extended separation distance **Dn** may refer to the separation distance **D** of the extended area **Se**. The extended area **Se** may be a concept including the aforementioned first area **S1** and second area **S2**. The extended separation distance **De** may be a concept including the aforementioned first separation distance **D1** and the second separation distance **D2**.

A recessed portion **346**, which is recessed radially inward, may be formed in the inner wall **344**. The recessed portion **346** may be referred to as a bent portion **346**. The bent portion **346** may be formed in a shape in which the inner wall **344** is pressed radially inward in the extended area **Se**. As the inner wall **344** is recessed radially inward, the separation distance **D** between the outer wall **342** and the inner wall **344** may increase.

Compared to the case where the outer wall **342** is bent outward to widen the passage of the diffuser **340**, the inner wall **344** is bent inward, thereby simply widening the passage of the diffuser **340** while maintaining a total volume of the diffuser **340**, such that an air volume of the blower **1** may increase.

The recessed portion **346** or the bent portion **345** may be formed by pressing the inner wall **344** in a radially inward direction. However, the technical idea of the present disclosure is not limited to the process of forming the recessed portion **346** or the bent portion **346**. For example, the recessed portion **346** or the bent portion **346** may also be formed by methods, such as thermal deformation, injection molding, and the like.

The recessed portion **346** may have a planar shape. In this case, the outer wall **342** may have a planar cross-section in a circular shape, and the inner wall **344** may be formed as a chord connecting a plurality of arcs to adjacent arcs.

Alternatively, the recessed portion **346** may be a curved surface which is convex inward (see FIG. **17**). As the bent portion **346** is formed as a curved surface that is convex inward, the width of the air passage **S** may further increase compared to the case where the bent portion **346** is simply formed as a planar surface.

As described above, the inner wall **344** of the diffuser **340** may share the load at the top of the blower **1**. In this case, the bent portion **346** (recessed portion **346**) which is bent inward is formed in the inner wall **344**, the load at the top of the blower **1** may be concentrated on the bent portion **346**. This is because, as a center of gravity of the blower **1** is positioned on the planar cross-section, a greater load is applied to a center portion, and thus the load is concentrated on the bent portion **345** of the inner wall **344**, which is bent inward and located in proximity to the center of gravity of the blower **1** compared to other portions.

Generally, the blower **1** may have an interior shape that is symmetrical with respect to the rotating shaft of the fan **320**.

15

or the motor shaft **312** of the fan motor **310**. Accordingly, the load applied to the inner wall **344** from the top of the blower **1** may be applied in a uniformly distributed manner on a virtual circumference around a central axis. Thus, when the inner wall **344** is recessed inward to secure a passage width of the diffuser **340**, there may be a problem in that the load is concentrated on the recessed portion **346** on the circumference CL.

The blower **1** may further include a sub-column **360** disposed within the inner wall **344**. The sub-column **360** may be disposed on the same circumference CL as the recessed portion **346** with respect to the center of the inner wall **344** (see FIG. 15).

As the sub-column **360** is disposed on the same circumference CL as the recessed portion **346**, the sub-column **360** and the recessed portion **346** may be spaced apart by the same radius from the center of the inner wall **344** (or the center of gravity of the blower **1** on the planar cross-section). Accordingly, the sub-column **360** and the recessed portion **346** may share the load applied from the top of the blower **1**.

Accordingly, the sub-column **360** disposed within the inner wall **344** and the recessed portion **346** of the inner wall **344** support the load at the top of the blower **1** on the same circumference CL, thereby preventing damage to the inner wall **344** when the load is concentrated on the recessed portion **346** of the inner wall **344**, and improving structural stability of the blower **1**.

A lower end of the sub-column **360** may be connected to the sub-part **344b** of the inner wall **344**. The sub-part **344b** of the inner wall **344** may support the lower end of the sub-column **360**. The sub-column **360** may have a pillar shape with a length which is the same as or similar to the main part **344a** of the inner wall **344**. An upper end of the sub-column **360** may be coupled to the lower portion of the supporter.

While the sub-column **360** is disposed to share the load as described above, the sub-column **360** may also be disposed in a shape that surrounds an outer circumference of the fan motor **310** to tightly secure the outer circumference of the fan motor **310**. It is obvious that by securing the fan motor **310**, vibrations and noise of the blower **1** may be reduced.

The plurality of vanes **348** may be spaced apart from each other around an outer circumference of the inner wall **344**. The plurality of vanes **348** may be spaced apart from each other around an inner circumference of the outer wall **342**.

The plurality of vanes **348** may be equally spaced apart around the circumference of the inner wall **344** or the outer wall **342** (see FIGS. 14 and 15). That is, unit passages ss divided by the vanes **348** may be spaced apart by equal separation distance D_v around the circumference. As the vanes **348** are spaced apart by equal separation distance D_v around the circumference, the vanes **348** may uniformly guide a flow direction regardless of an area of the unit passages ss.

By contrast, the plurality of vanes **348** may not be equally spaced apart. The separation distance D_v between adjacent vanes **348** of the plurality of vanes **348** in the extended area S_e may be greater than a normal area S_n (see FIG. 16). The separation distance D_v between the vanes **348** may be understood as a separation distance between the vanes **348** on the circumference. That is, the number of vanes **348** disposed in the normal area S_n may be greater than the number of vanes **348** disposed in the extended region S_e .

Accordingly, while the vanes **348** may serve to guide a flow in an upward direction, the vanes **348** may also locally increase flow resistance. Thus, an air volume in a specific

16

area may be improved by reducing the number of vanes **348** in an area in which it is required to secure the air volume.

A plurality of grooves (not shown) may be formed circumferentially in an inner surface of the outer wall **342** and an outer surface of the inner wall **344**. Separation distances between adjacent grooves of the plurality of grooves may be equally spaced apart. The plurality of vanes **348** may be press-fitted into predetermined grooves of the plurality of grooves, to be fastened thereto.

A separation distance between the vanes **348** in the extended region S_e may be adjusted simply by allowing the vanes **348** to be selectively fastened to the grooves formed at equal intervals. In addition, when an internal structure of the blower **1** is changed by adding an option and the like during use of the blower **1**, it is possible to simply adjust the blower according to the changed structure.

Hereinafter, the upper case **140**, the blowing space **105**, and a flow of discharged air will be described with reference to FIGS. 1 to 4.

As described above, the upper case **140** may include the first tower **110** and the second tower **120** which are separately disposed and are in the shape of two pillars. In this embodiment, the first tower **110** may be disposed on the left side, and the second tower **120** may be disposed on the right side. The first tower **110** and the second tower **120** may be separated from each other, and the blowing space **105** may be formed between the first tower **110** and the second tower **120**.

In this embodiment, the blower space **105** may be open at the front, rear, and/or the top.

The upper case **140** including the first tower **110**, the second tower **120**, and the blowing space **105** may be formed in a truncated cone shape.

The discharge ports **117** and **127** formed in the first tower **110** and the second tower **120**, respectively, may discharge air into the blowing space **105**. If it is required to distinguish the discharge ports **117** and **127**, one of the discharge ports **117** and **127**, which is formed in the first tower **110**, may be referred to as the first discharge port **117**, and the other one of the discharge ports **117** and **127**, which is formed in the second tower **120**, may be referred to as the second discharge port **127**.

The first and second discharge ports **117** and **127** may be disposed within the height of the blowing space **105**, and a transverse direction across the blowing space **105** is defined as an air discharge direction.

As the first tower **110** and the second tower **120** are disposed on the left and right sides, the air discharge direction may be formed in a front-rear direction and an up-down direction in this embodiment.

That is, the air discharge direction across the blowing space **105** may include a first air discharge direction $F1$ that is horizontally oriented and a second air discharge direction $F2$ that is vertically oriented.

Air flowing in the first air discharge direction $F1$ is referred to as a horizontal airflow, and air flowing in the second air discharge direction $F2$ is referred to as an ascending airflow.

It should be understood that the horizontal airflow does not mean that the air flows only in the horizontal direction, but that a flow rate of air flowing in the horizontal direction is larger. Likewise, it should be understood that the ascending airflow does not mean that the air flows only upward, but that a flow rate of air flowing upward is larger.

In this embodiment, a gap between upper ends of the blowing space **105** may be equal to a gap between lower ends thereof. Unlike this embodiment, the gap between the

17

upper ends of the blowing space **105** may also be narrower or wider than the gap between the lower ends thereof.

As the blowing space **105** has a uniform horizontal width, a flow of air flowing in front of the blowing space **105** may be formed more uniformly.

For example, when a width of the upper side and a width of the lower side are different, a flow velocity of the wider side may be formed low, and a deviation of the velocities may occur based on the up-down direction. When the velocity deviation of the air occurs in the up-down direction, an air reaching length may vary.

The air discharged from the first discharge port **117** and the air discharged from the second discharge port **127** may join each other in the blowing space **105** to flow to a user.

That is, in this embodiment, without individually flowing to the user, the air discharged from the first discharge port **117** and the air discharged from the second discharge port **127** may join in the blowing space **105**, and then is provided to the user.

The blowing space **105** may be used as a space in which the discharged air flows join each other and are mixed. In addition, air behind the blowing space **105** may also flow into the blowing space **105** by the air discharged into the blowing space **105**.

As the air discharged from the first discharge port **117** and the air discharged from the second discharge port **127** join in the blowing space **105**, straightness of the discharged air may be improved. In addition, by joining the air discharged from the first discharge port **117** and the air discharged from the second discharge port **127** in the blowing space **105**, air around the first tower and second tower can also indirectly flow in the air discharge direction.

In this embodiment, the first air discharge direction **F1** may be oriented from rear to front, and the second air discharge direction **F2** may be oriented from bottom to top.

An upper end **111** of the first tower **110** and an upper end **121** of the second tower **120** may be spaced apart from each other for the second air discharge direction **F2**. That is, the air discharged in the second air discharge direction **F2** causes no interference with the case **100** of the blower **1**.

Further, for the first air discharge direction **F1**, a front end **112** of the first tower **110** and a front end **122** of the second tower **120** may be spaced apart from each other, and a rear end **113** of the first tower **110** and a rear end **123** of the second tower **120** may also be spaced apart from each other.

In each of the first tower **110** and the second tower **120**, a surface facing the blowing space **105** is referred to as an inner wall, and a surface not facing the blowing space **105** is referred to as an outer wall.

An outer wall **114** of the first tower **110** and an outer wall **124** of the second tower **120** may be disposed in opposite directions, and an inner wall **115** of the first tower **110** and an inner wall **125** of the second tower **120** may face each other.

If it is necessary to distinguish the inner walls **115** and **125**, the inner wall of the first tower **110** is referred to as the first inner wall **115**, and the inner wall of the second tower **120** is referred to as the second inner wall **125**.

Likewise, if it is necessary to distinguish the outer walls **114** and **124**, the outer wall of the first tower **110** is referred to as the first outer wall **114**, and the outer wall of the second tower **120** is referred to as the second outer wall **124**.

The first tower **110** and the second tower **120** may be formed in a streamlined shape with respect to the flow direction of air.

Specifically, each of the first inner wall **115** and the first outer wall **114** is formed in a streamlined shape in the

18

front-rear direction, and each of the second inner wall **125** and the second outer wall **124** is formed in a streamlined shape in the front-rear direction.

The first discharge port **117** is formed in the first inner wall **115**, and the second discharge port **127** is formed in the second inner wall **125**.

A shortest distance between the first inner wall **115** and the second inner wall **125** is referred to as **B0**. The discharge ports **117** and **127** may be disposed behind the shortest distance **B0**.

A separation distance between the front end **112** of the first tower **110** and the front end **122** of the second tower **120** is referred to as a first separation distance **B1**, and a separation distance between the rear end **113** of the first tower **110** and the rear end **123** of the second tower **120** is referred to as a second separation distance **B2**.

In this embodiment, **B1** and **B2** may be equal to each other. Unlike this embodiment, any one of **B1** or **B2** may be longer than the other.

The first discharge port **117** and the second discharge port **127** may be formed between **B0** and **B2**.

Preferably, the first discharge port **117** and the second discharge port **127** are disposed closer to the rear end **113** of the first tower **110** and the rear end **123** of the second tower **120** than **B0**. As the discharge ports **117** and **127** are disposed closer to the rear ends **113** and **123**, air flow may be easily controlled by the Coanda effect.

The inner wall **115** of the first tower **110** and the inner wall **125** of the second tower **120** may directly provide the Coanda effect, and the outer wall **114** of the first tower **110** and the outer wall **124** of the second tower **120** may indirectly provide the Coanda effect.

The inner walls **115** and **125** may directly guide the air discharged from the discharge ports **117** and **127** to the front ends **112** and **122**.

That is, the horizontal airflow of the air discharged from the discharge ports **117** and **127** may be directly provided.

Due to an air flow in the blowing space **105**, an indirect air flow occurs in the outer walls **114** and **124**.

The outer walls **114** and **124** may induce a Coanda effect with respect to the indirect air flow, and may guide the indirect air flow to the front ends **112** and **122**.

A left side of the blowing space **105** is blocked by the first inner wall **115**, and a right side of the blowing space **105** is blocked by the second inner wall **125**, but an upper side of the blowing space **105** may be opened.

An airflow converter to be described later may convert the horizontal airflow passing through the blowing space **105** into the ascending airflow, and the ascending airflow may flow to the open upper side of the blowing space **105**. The ascending airflow suppresses the direct flow of discharged air to the user and may allow for active convection of indoor air.

In addition, a width of discharged air may be adjusted using the flow rate of air that joins in the blowing space **105**.

By having a vertical length of the first discharge port **117** and the second discharge port **127** which are much longer than the horizontal widths **B0**, **B1**, and **B2** of the blowing space **105**, the discharged air of the first discharge port **117** and the discharged air of the second discharge port **127** may be caused to join each other in the blowing space **105**.

Hereinafter, the overall shape of the blower **1** will be described with reference to FIGS. **1** to **4**.

The blower **1** may have a pillar shape having a diameter that decrease upward. The blower **1** may have a substantially cone shape or a truncated cone shape. In this embodiment, the lower case **150** may have a diameter that gradually

19

decreases toward the upper end. The middle case **130** may have a diameter that gradually decreases toward the upper end. The upper case **140** may also have a diameter that gradually decreases toward the upper end.

Unlike this embodiment, the blower **1** may include a form in which two towers are disposed. In addition, unlike this embodiment, it is not necessary to have a shape with a cross-section that becomes narrower toward the top.

However, as shown in this embodiment, if the cross-section becomes narrower toward the top, the center of gravity is lowered and a risk of inversion due to an external force is reduced.

Unlike this embodiment, the lower case **150**, the middle case **130**, and the upper case **140** may be integrally formed with each other. For example, the lower case **150**, the middle case **130**, and the upper case **140** may be manufactured in the form of a front case and a rear case which are integrally formed, and then assembled with each other.

The outer surfaces of the lower case **150**, the middle case **130**, and the upper case **140** may be formed continuously. Particularly, a lower end of the tower base **130** and an upper end of the lower case **150** may be in close contact with each other, and the outer surface of the tower base **130** and the outer surface of the lower case **150** may form a continuous surface. To this end, a diameter of the lower end of the tower base **130** may be slightly smaller than a diameter of the upper end of the lower case **150**.

The lower case **150** may have an inner space and the suction port **155** formed in a circumference thereof. The lower case **150** may further include a base **151** which is seated on the ground. The lower case **150** may be separated laterally to provide a path for removing or installing a filter. In this embodiment, the suction port **155** may be formed along the circumference of the lower case **150**, thereby suctioning air in all directions 360 degrees. In this embodiment, the suction port **155** may be formed in the shape of a hole, and may have various shapes.

The tower base **130** may distribute air supplied from the lower case **150**, and may provide the distributed air to the first tower **110** and the second tower **120**. The tower base **130** may connect the first tower **110** and the second tower **120**. The blowing space **105** may be disposed at an upper side of the tower base **130**. In addition, the discharge ports **117** and **127** may be formed at the upper side of the tower base **130**, and the ascending airflow and the horizontal airflow may be formed at the upper side of the tower base **130**.

In order to minimize a friction with air, an upper surface **131** of the tower base **130** may be formed as a curved surface. Particularly, the upper surface **131** may be formed as a curved surface which is concave downward, and may extend in the front-rear direction. A first side **131a** of the upper surface **131** may be connected to the first inner wall **115**, and a second side **131b** of the upper surface **131** may be connected to the second inner wall **125**. The upper surface **131**, the first inner wall **115**, and the second inner wall **125** may form a continuous surface. The upper surface **131** of the tower base **130** may be referred to as a bridge surface **131**.

Referring to FIG. 4, when viewed from the top, the first tower **110** and the second tower **120** may be symmetrical to each other with respect to a center line L-L'. Particularly, the first discharge port **117** and the second discharge port **127** may be symmetrical to each other with respect to the center line L-L'.

The center line L-L' is an imaginary line between the first tower **110** and the second tower **120** and is disposed in a

20

front-rear direction in this embodiment, and is disposed to pass through the upper surface **131**.

Unlike this embodiment, it is also possible that the first tower **110** and the second tower **120** may be formed in an asymmetric shape. However, when the first tower **110** and the second tower **120** are disposed symmetrical with respect to the center line L-L', it is more advantageous to control horizontal airflow and ascending airflow.

Referring to FIGS. 5 to 7, the blower **1** may include a filter **200** disposed in the case **100**.

The filter **200** may be disposed in the lower case **150**. The lower case **150** is removably disposed to provide a path for removing or installing the filter. A user may remove the lower case **150** to withdraw the filter **200** out of the case **100**.

The filter **200** may be formed in a cylindrical shape which is hollow in the up-down direction therein. An outer surface of the filter **200** may be disposed opposite the suction port **155**. Indoor air may pass through the filter from the outside toward the inside, and foreign substances or harmful gases may be removed from the air.

The fan will be described below with reference to FIGS. 5 to 7 and 13.

The fan **320** may be disposed in the lower case **150**. The fan **320** may be disposed at an upper side of the filter **200**. The fan **320** may cause the air, having passed through the filter **200**, to flow to the first tower **110** and the second tower **120**. The fan **320** may be rotated by the fan motor **310**.

The fan motor **310** may be disposed above the fan **320**, and the motor shaft **312** of the fan motor **310** may be coupled to the fan **320** disposed at a lower side thereof.

The fan motor **310** may be accommodated by the inner wall **344** of the diffuser. The inner wall **344** may have a shape that covers the entire fan motor **310**. As the inner wall **344** covers the entire fan motor **310**, flow resistance of air flowing from bottom to top may be reduced. In this case, in order to minimize volume in an up-down direction, a lower end of the inner wall **344** may be inserted into the fan **320** and may overlap with the fan **320**.

However, as described above, the inner wall **344** of the diffuser accommodates the fan motor **310** merely for space efficiency, and it should not be construed as limiting the technical idea of the present disclosure.

The fan **320** may include a hub **322**, to which the shaft of the fan motor **310** is coupled, a shroud **324** spaced apart from the hub, and a plurality of blades **326** connecting the hub and the shroud.

Air having passed through the filter **200** is drawn into the shroud, and then is pressurized by the rotating blade **326** to cause the air to flow. The hub **322** is disposed at the upper side of the blade **326**, and the shroud **324** is disposed under the blade **326**. The hub **322** may be formed in a bowl shape that is concave downward. A width of the discharge portion of the fan **320** may refer to a distance between the hub **322** and the shroud **324**.

In this embodiment, a mixed flow fan may be used as the fan **320**. The mixed flow fan has a characteristic of sucking air into an axial center, and radially discharging air, in which the discharged air is inclined with respect to the axial direction. The entire air flows from the lower side to the upper side, such that when air is discharged in the radial direction as in a general centrifugal fan, a large flow loss occurs due to the flow direction change. However, the technical idea of the present disclosure is not limited to the type of fan.

Meanwhile, a suction grill **350** may be disposed in the lower case **150**. When the filter **200** is removed, the suction

grill **350** prevents a finger of the user from entering the fan **320**, thereby protecting the user and the fan **320**.

The filter **200** may be disposed at a lower side of the suction grill **350**, and the fan **320** may be disposed at an upper side thereof. The suction grill **350** has a plurality of through holes formed in the up-down direction so that air may flow.

Referring to FIGS. **8** and **9**, the first discharge port **117** of the first tower **110** may be directed toward the second tower **120**, and the second discharge port **127** of the second tower **120** may be directed toward the first tower **110**.

The air discharged from the first discharge port **117** may flow along the inner wall **115** of the first tower **110** by the Coanda effect. The air discharged from the second discharge port **127** may flow along the inner wall **125** of the second tower **120** by the Coanda effect.

The first discharge part **170** forms the first discharge port **117**, and includes a first discharge guide **172** which is disposed on an air discharge side of the first discharge port **117**, and a second discharge guide **174** which forms the first discharge port **117** and is disposed on a side opposite the air discharge side of the first discharge port **117**.

Outer surfaces **172a** and **174a** of the first discharge guide **172** and the second discharge guide **174** may provide a portion of the inner wall **115** of the first tower **110**.

An inside of the first discharge guide **172** may be disposed to face the first discharge passage **O1**, and an outside thereof may be disposed to face the blowing space **105**. An inside of the second discharge guide **174** may be disposed to face the first discharge passage **O1**, and an outside thereof may be disposed to face the blowing space **105**.

An outer surface **172a** of the first discharge guide **172** may be formed as a curved surface. The outer surface **172a** may provide a continuous surface with the first inner wall **115**. Particularly, the outer surface **172a** may form a continuous curved surface with an outer surface of the first inner wall **115**.

An outer surface **174a** of the second discharge guide **174** may provide a continuous surface with the first inner wall **115**. The inner surface **174b** of the second discharge guide **174** may be formed as a curved surface. Particularly, the inner surface **174b** may form a continuous curved surface with an inner surface of the first outer wall **114**, thereby guiding air in the first discharge passage **O1** toward the first discharge guide **172**.

The first discharge port **117** may be formed between the first discharge guide **172** and the second discharge guide **174**, and air in the first discharge passage **O1** may be discharged into the blowing space **105** through the first discharge port **117**.

Specifically, the air in the first discharge passage **O1** may be discharged between the outer surface **172a** of the first discharge guide **172** and the inner surface **174b** of the second discharge guide **174**, and a space between the outer surface **172a** of the first discharge guide **172** and the inner surface **174b** of the second discharge guide **174** is defined as a discharge gap **175**. The discharge gap **175** forms a predetermined channel.

The discharge gap **175** is formed so that a width of the intermediate portion **175b** is narrower than those of an inlet **175a** and an outlet **175c**. The intermediate portion **175b** is defined as the shortest distance in the discharge gap **175**.

A cross-sectional area of the discharge gap **175** gradually decreases from the inlet of the discharge gap **175** to the intermediate portion **175b** thereof, and the cross-sectional area gradually increases again from the intermediate portion **175b** to the outlet **175c**. The intermediate portion **175b** may

be located inside the first tower **110**. When viewed from the outside, the outlet **175c** of the discharge gap **175** may be viewed as the discharge port **117**.

In order to induce the Coanda effect, a radius of curvature of the inner surface **174b** of the second discharge guide **174** may be greater than a radius of curvature of the outer surface **172a** of the first discharge guide **172**.

A center of curvature of the outer surface **172a** of the first discharge guide **172** is located in front of the outer surface **172a** and is formed inside the first discharge passage **O1**. A center of curvature of the inner surface **174b** of the second discharge guide **174** is located on the side of the first discharge guide **172** and is formed inside the first discharge passage **O1**.

The second discharge part **180** forms the second discharge port **127**, and includes a first discharge guide **182** which is disposed on an air discharge side of the second discharge port **127**, and a second discharge guide **184** which forms the second discharge port **127** and is disposed on a side opposite the air discharge side of the second discharge port **127**.

A discharge gap **185** may be formed between the first discharge guide **182** and the second discharge guide **184**. The second discharge part **180** is symmetrical with the first discharge part **170**, such that a detailed description thereof will be omitted.

The blower **1** may further include an airflow converter **400** for changing an air flow direction of the blowing space **105**.

FIG. **10** is a horizontal cross-sectional view of an airflow converter of FIG. **2**. The airflow converter **400**, capable of generating an ascending airflow, will be described with reference to FIG. **7** or FIG. **16**.

In this embodiment, the airflow converter **400** may convert a horizontal flow of air, flowing through the blowing space **105**, into an ascending airflow.

The airflow converter **400** may include a first airflow converter **401** disposed in the first tower **110**, and a second airflow converter **402** disposed in the second tower **120**. The first airflow converter **401** and the second airflow converter **402** are symmetrical with each other and have the same configuration.

The airflow converter **400** may include a guide board **410** disposed in the tower and protruding into the blowing space **105**, a guide motor **420** providing a driving force for moving the guide board **410**, a power transmission member **430** providing the driving force of the guide motor **420** to the guide board **410**, and a board guider **440** disposed inside the tower and guiding the movement of the guide board **410**.

The guide board **410** may be hidden inside the tower and may protrude into the blowing space **105** when the guide motor **420** is activated.

In this embodiment, the guide board **410** includes a first guide board **411** disposed in the first tower **110**, and a second guide board **412** disposed in the second tower **120**.

To this end, a board slit **119** passing through the inner wall **115** of the first tower **110** and a board slit **129** passing through the inner wall **125** of the second tower **120** may be formed, respectively.

The board slit **119** formed in the first tower **110** is referred to as a first board slit **119**, and the board slit formed in the second tower **120** is referred to as a second board slit **129**.

The first board slit **119** and the second board slit **129** may be symmetrical with each other. The first board slit **119** and the second board slit **129** may be elongated in an up-down direction. The first board slit **119** and the second board slit **129** may be inclined with respect to a vertical direction **V**.

With respect to the vertical direction forming an angle of zero degrees, the front end **112** of the first tower **110** may be formed to have a first inclination and the first board slit **119** may be formed to have a second inclination. The front end **122** of the second tower **120** may also be formed to have the first inclination and the second board slit **129** may also be formed to have the second inclination.

The first inclination may range between the vertical direction and the second inclination, and the second inclination may be greater than a horizontal direction. The first inclination and the second inclination may be equal to each other, or the second inclination may be greater than the first inclination.

The guide board **410** may be formed in a flat plate or curved plate shape. The guide board **410** may be elongated in the up-down direction, and may be disposed at the front of the blowing space **105**.

The guide board **410** may block the horizontal flow of air, flowing into the blowing space **105**, to change the direction of airflow to an upward direction.

In this embodiment, an inner end **411a** of the first guide board **411** and an inner end **412a** of the second guide board **412** may come into contact with or abut on each other to form an ascending airflow. Unlike this embodiment, one guide board **410** may be in close contact with an opposite tower to form an ascending airflow.

When the airflow converter **400** is not in operation, the inner end **411a** of the first guide board **411** may close the first board slit **119**, and the inner end **412a** of the second guide board **412** may close the second board slit **129**.

When the airflow converter **400** is activated, the inner end **411a** of the first guide board **411** may pass through the first board slit **119** to protrude into the blowing space **105**, and the inner end **412a** of the second guide board **412** may pass through the second board slit **129** to protrude into the blowing space **105**.

In this embodiment, the first guide board **411** and the second guide board **412** may protrude into the blowing space **105** by rotation. Unlike this embodiment, it is also possible that at least one of the first guide board **411** and the second guide board **412** may linearly move by sliding to protrude into the blowing space **105**.

When viewed from the top, the first guide board **411** and the second guide board **412** may be formed in an arc shape. Each of the first guide board **411** and the second guide board **412** may form a predetermined radius of curvature, and a center of curvature of each of the first guide board **411** and the second guide board **412** may be located in the blowing space **105**.

When the guide board **410** is hidden inside the tower, a volume of the guide board **410** in a radially inward direction is preferably greater than a volume of the guide board **410** in a radially outward direction.

The guide board **410** may be made of a transparent material. A light-emitting member, such as a light-emitting diode (LED), may be disposed at the guide board **410**, and light emitted by the light-emitting member may allow the entire guide board **410** to be illuminated. The light-emitting member may be disposed in the discharge passage **O** in the tower and may be disposed at an outer end of the guide board **410**.

The guide motor **420** may include a first guide motor **421** providing torque to the first guide board **411**, and a second guide motor **422** providing torque to the second guide board **412**.

The first guide motor **421** may be disposed on each of an upper side and a lower side in the first tower **110**, and if it

is necessary to distinguish the first guide motor, the first guide motor may be divided into an upper first guide motor **421** and a lower first guide motor **421**. The upper first guide motor is disposed below the upper end **111** of the first tower **110**, and the lower first guide motor is disposed above the fan **320**.

The second guide motor **422** may also be disposed on each of an upper side and a lower side in the second tower **120**, and if it is necessary to distinguish the second guide motor, the second guide motor may be divided into an upper second guide motor **422a** and a lower second guide motor **422b**. The upper second guide motor is disposed below the upper end **121** of the second tower **120**, and the lower second guide motor is disposed above the fan **320**.

In this embodiment, rotating shafts of the first guide motor **421** and the second guide motor **422** may be disposed vertically, and a rack-pinion structure may be used to transmit a driving force.

The power transmission member **430** may include a driving gear **431** coupled to the motor shaft **312** of the guide motor **420**, and a rack **432** coupled to the guide board **410**.

The driving gear **431** is a pinion gear and may rotate horizontally.

The rack **42** may be coupled to an inner surface of the guide board **410**. The rack **432** may be disposed in the discharge passage **O**, and may perform a turning motion with the guide board **410**.

The board guider **440** may guide the turning motion of the guide board **410**. The board guider **440** may support the guide board **410** when the guide board **410** performs the turning motion.

In this embodiment, the board guider **440** may be disposed on an opposite side of the rack **432** with respect to the guide board **410**. The board guider **440** may support a force applied by the rack **432**. Unlike this embodiment, it is also possible that a groove corresponding to a turning radius of the guide board may be formed in the board guider **440**, and the guide board may be moved along the groove.

The board guider **440** may be assembled to the outer walls **114** and **124** of the tower. The board guider **440** may be disposed radially outward from the guide board **410**, thereby minimizing contact with the air flowing through the discharge passage **O**.

FIG. **11** is a diagram illustrating an example of a horizontal airflow of a blower according to an embodiment of the present disclosure.

Referring to FIG. **11**, when the horizontal airflow is provided, the first guide board **411** is hidden inside the first tower **110**, and the second guide board **412** is hidden inside the second tower **120**.

The air discharged from the first discharge port **117** and the air discharged from the second discharge port **127** may join in the blowing spaces **105** and **120**, and may pass through the front ends **112** and **122** to flow forward.

Further, the air behind the blowing space **105** may be guided into the blowing space **105**, to flow forward.

In addition, the air around the first tower **110** may flow forward along the first outer wall **114**, and the air around the second tower **120** may flow forward along the second outer wall **124**.

The first discharge port **117** and the second discharge port **127** are elongated in the up-down direction and are symmetrical with each other, thereby uniformly distributing the air flowing along the upper side of the first discharge port **117** and the second discharge port **127** and the air flowing along the lower side thereof.

25

In addition, the air discharged from the first and second discharge ports 117 and 127 join in the blowing space 105, thereby improving straightness of the discharged air and allowing the air to flow a longer distance.

FIG. 12 is a diagram illustrating an example of an ascending airflow of a blower according to an embodiment of the present disclosure.

Referring to FIG. 12, when an ascending airflow is provided, the first guide board 411 and the second guide board 412 protrude into the blowing space 105 and may block a front side of the blowing space 105.

As the front side of the blowing space 105 is blocked by the first guide board 411 and the second guide board 412, the air discharged from the discharge ports 117 and 127 may flow upward along the rear surfaces of the first guide board 411 and the second guide board 412 and is discharged to the top of the blowing space 105.

By forming the ascending airflow in the blower 1, it is possible to prevent the discharged air from flowing directly to the user. In addition, the indoor air may be circulated by operating the blower 1 in the ascending airflow mode.

For example, by operating the blower 1 in the ascending airflow mode when the air conditioner and the blower are used at the same time, convection of the indoor air may be promoted, and the indoor air may be cooled or heated more quickly.

The above description of the present disclosure is provided for the purpose of illustration, and it would be understood by those skilled in the art, to which the present disclosure pertains, that various changes and modifications may be made without changing technical conception and essential features of the present disclosure. Thus, it is clear that the above-described embodiments are illustrative in all aspects and are not intended to limit the present disclosure. The scope of the present disclosure is defined by the following claims rather than by the detailed description of the embodiment. It shall be understood that all modifications and embodiments conceived from the meaning and scope of the claims and their equivalents are included in the scope of the present disclosure.

What is claimed is:

1. A blower, comprising:

a case having a suction port and discharge ports;

a fan disposed inside of the case; and

a diffuser that is disposed downstream of the fan and guides air, discharged from the fan, toward the discharge ports, wherein the diffuser comprises:

an outer wall having a circumference;

an inner wall spaced apart inwardly from the outer wall so that an air passage, through which the air discharged from the fan flows, is formed therebetween; and

a plurality of vanes that extends radially between the outer wall and the inner wall and divides the air passage into a plurality of unit passages, wherein the outer wall and the inner wall are formed to have areas with different radial separation distances D-between the outer wall and the inner wall, wherein the air passage comprises a normal area in which the separation distance is a predetermined value, and an extended area in which the separation distance is greater than the predetermined value, wherein a plurality of recessed portions, which is recessed radially inward, is formed in the inner wall of the extended area, and wherein the plurality of recessed portions is spaced apart from each other along a circumferential direction of the inner wall.

26

2. The blower of claim 1, wherein the case comprises:

a lower case having the suction port formed therein, and the fan and the diffuser disposed therein;

an upper case having the discharge ports formed therein, disposed over the lower case, and having a discharge passage disposed therein through which the air flows toward the discharge ports; and

a middle case disposed between the lower case and the upper case, and having an intermediate passage to allow the air passage to communicate with the discharge passage.

3. The blower of claim 2, further comprising a penetrating member having at least a portion that penetrates into the middle case and disposed in the intermediate passage, wherein the separation distance in a first area is larger than in an area adjacent to the first area in the air passage of the diffuser, the first area being defined as a projected area in which the penetrating member is projected to the air passage of the diffuser.

4. The blower of claim 2, wherein the upper case comprises:

a first discharge case having a first discharge port and a first discharge passage through which the air flows toward the first discharge port; and

a second discharge case having a second discharge port and a second discharge passage through which the air flows toward the second discharge port, the second discharge case being spaced apart from the first discharge case such that a space is formed therebetween, wherein each of the first discharge passage and the second discharge passage communicates with the intermediate passage of the middle case, and wherein the separation distance in at least a portion of a second area is formed to be larger than in an area adjacent to the second area in the air passage of the diffuser, the second area being defined as an area in which the first discharge passage and the second discharge passage are projected to the air passage.

5. The blower of claim 4, wherein the first discharge case and the second discharge case are symmetrical to each other with respect to the space formed therebetween, and wherein distribution of the separation distance is symmetrical with respect to a third area, which is an area in which the space is projected to the air passage.

6. The blower of claim 1, wherein the blower further comprises:

a first tower disposed above the diffuser, having a first discharge port of the discharge ports, and elongated upward;

a second tower disposed above the diffuser, having a second discharge port of the discharge ports, elongated upward, and laterally spaced apart from the first tower such that a space is formed therebetween;

a bridge surface that connects lower ends of side walls of the first tower and the second tower which face each other with respect to the space; and

a supporter having a lower portion connected to an upper end of an inner wall and an upper portion connected to a lower side of the bridge surface, so as to transfer load from the bridge surface to the inner wall, wherein the plurality of vanes have radially inner ends connected to the inner wall and radially outer ends connected to the outer wall.

7. The blower of claim 6, wherein the diffuser further comprises at least one sub-column disposed within the inner wall, disposed on a same circumference as the plurality of

27

recessed portions with respect to a center of the inner wall, and having an upper end connected to the lower portion of the supporter.

8. The blower of claim 1, wherein the plurality of vanes is equally spaced apart around the circumference.

9. The blower of claim 1, wherein a separation distance between adjacent vanes of the plurality of vanes is larger in the extended area than in the normal area.

10. The blower of claim 9, wherein a plurality of grooves is formed circumferentially in an inner surface of the outer wall and an outer surface of the inner wall, and wherein: adjacent grooves among the plurality of grooves are equally spaced apart; and

the plurality of vanes are press-fit into predetermined grooves of the plurality of grooves, to be fastened thereto.

11. The blower of claim 1, wherein the inner wall has a bent portion pressed radially inward in the extended area.

12. The blower of claim 11, wherein the bent portion is a curved surface that is convex inward.

13. A blower, comprising:

a case having a suction port and discharge ports;

a fan disposed inside of the case; and

a diffuser that is disposed downstream of the fan and guides air, discharged from the fan, toward the discharge ports, wherein the diffuser comprises:

an outer wall having a circumference;

an inner wall spaced apart inwardly from the outer wall so that an air passage, through which the air discharged from the fan flows, is formed therebetween; and

a plurality of vanes that extends radially between the outer wall and the inner wall and divides the air passage into a plurality of unit passages, wherein the outer wall and the inner wall are formed to have areas with different radial separation distances between the outer wall and the inner wall, wherein the air passage comprises a normal area in which the separation distance is a predetermined value, and an extended area in which the separation distance is greater than the predetermined value, and wherein the inner wall has at least one bent portion pressed radially inward in the extended area.

14. The blower of claim 13, wherein the case comprises: a lower case having the suction port formed therein, and the fan and the diffuser disposed therein;

an upper case having the discharge ports formed therein, disposed over the lower case, and having a discharge passage disposed therein through which the air flows toward the discharge ports; and

a middle case disposed between the lower case and the upper case, and having an intermediate passage to allow the air passage to communicate with the discharge passage.

15. The blower of claim 13, wherein the blower further comprises:

a first tower disposed above the diffuser, having a first discharge port of the discharge ports, and elongated upward;

a second tower disposed above the diffuser, having a second discharge port of the discharge ports, elongated upward, and laterally spaced apart from the first tower such that a space is formed therebetween;

a bridge surface that connects lower ends of side walls of the first tower and the second tower which face each other with respect to the space; and

28

a supporter having a lower portion connected to an upper end of an inner wall and an upper portion connected to a lower side of the bridge surface, so as to transfer load from the bridge surface to the inner wall, wherein the plurality of vanes have radially inner ends connected to the inner wall and radially outer ends connected to the outer wall.

16. The blower of claim 13, wherein a separation distance between adjacent vanes of the plurality of vanes is larger in the extended area than in the normal area.

17. A blower, comprising:

a case having a suction port and discharge ports;

a fan disposed inside of the case; and

a diffuser that is disposed downstream of the fan and guides air, discharged from the fan, toward the discharge ports, wherein the diffuser comprises:

an outer wall having a circumference;

an inner wall spaced apart inwardly from the outer wall so that an air passage, through which the air discharged from the fan flows, is formed therebetween;

a plurality of vanes that extends radially between the outer wall and the inner wall and divides the air passage into a plurality of unit passages, wherein the outer wall and the inner wall are formed to have areas with different radial separation distances between the outer wall and the inner wall, wherein the air passage comprises a normal area in which the separation distance is a predetermined value, and an extended area in which the separation distance is greater than the predetermined value; and

at least one sub-column disposed within the inner wall and having an upper end connected to a lower portion of a supporter.

18. The blower of claim 17, wherein the case comprises: a lower case having the suction port formed therein, and the fan and the diffuser disposed therein;

an upper case having the discharge ports formed therein, disposed over the lower case, and having a discharge passage disposed therein through which the air flows toward the discharge ports; and

a middle case disposed between the lower case and the upper case, and having an intermediate passage to allow the air passage to communicate with the discharge passage.

19. The blower of claim 17, wherein the blower further comprises:

a first tower disposed above the diffuser, having a first discharge port of the discharge ports, and elongated upward;

a second tower disposed above the diffuser, having a second discharge port of the discharge ports, elongated upward, and laterally spaced apart from the first tower such that a space is formed therebetween;

a bridge surface that connects lower ends of side walls of the first tower and the second tower which face each other with respect to the space; and

the supporter having the lower portion of which is connected to the upper end of the inner wall and an upper portion of which is connected to a lower side of the bridge surface, so as to transfer load from the bridge surface to the inner wall, wherein the plurality of vanes have radially inner ends connected to the inner wall and radially outer ends connected to the outer wall.

20. The blower of claim 17, wherein a separation distance between adjacent vanes of the plurality of vanes is larger in the extended area than in the normal area.